

The Cost of Regulatory Compliance in the United States

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A key question for studying business dynamism is whether the costs of regulatory compliance fall homogeneously on small and large businesses. Using comprehensive establishment-level occupational microdata and occupation task information, we quantify a firm's compliance costs as the share of wage bill for performing regulatory compliance tasks (RegIndex). We reveal an inverted-U relationship between firms' RegIndex and their size: On average, RegIndex for mid-sized firms with around 500 employees is about 47% greater than that of the smallest firms and 18% greater than that of the largest firms. We further develop a shift-share methodology to disentangle the influence of regulatory requirements and enforcement on driving firms' compliance costs. (JEL G38, J01, K2, L43, L51)

We would like to thank the editor, the associate editor, and two anonymous referees for helpful suggestions. We also thank Matt Backus, Andrew Baker, Matilde Bombardini, Greg Buchak, Julieta Caunedo, Ernesto Dal Bó, Steven Davis, Klaus Desmet, Fred Finan, Bård Harstad, Joseph Kalmenovitz, Alessandro Lizzeri, John Matsusaka, Adair Morse, Tom Palfrey, Amit Seru, Steven Davidoff Solomon, James Spindler, Reed Walker, Jessie Wang, Joshua White, and seminar participants at the NBER Corporate Finance Fall Meeting 2023, NBER Economic Analysis of Regulation Conference, NBER SI Law and Economics 2024, Tepper-LAEF Conference, WFA, PSL Public Governance Seminar, Berkeley Political Economy Research Lab, Arizona State University, Northwestern University, Stanford GSB, University of California College of Law, University of Michigan, University of Southern California, University of Texas at Austin, Vanderbilt University, and other institutions for their comments. Niranjana Navaneethan provided excellent research assistance. This research was conducted with restricted access to Bureau of Labor Statistics data. The views expressed here do not necessarily reflect the views of the BLS. We thank Erin Good, Jessica Helfand, Michael Horrigan, Mark Loewenstein, and especially Michael Soloy at the BLS for their excellent assistance. A separate, unpublished manuscript, "Regulation Intensity and Technology-Driven Entry" by Michael Simkovic and Miao Ben Zhang, which described some of the aggregate public BLS data, is now retired. [Supplementary data](#) can be found on *The Review of Financial Studies* web site. Send correspondence to Miao Ben Zhang, miao.zhang@marshall.usc.edu.

The Review of Financial Studies 39 (2026) 2901–2962

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<https://doi.org/10.1093/rfs/hhag046>

Advance Access publication April 29, 2026

Received: October 24, 2024

Editor: Manju Puri

Authors have furnished an [Internet Appendix](#), which is available on the Oxford University Press Web site next to the link to the final published paper online.

Regulation is frequently ascribed as one of the putative drivers for rising industry concentration and underinvestment in the United States (Gutiérrez and Philippon 2019, 2017) and in other countries (Aghion, Bergeaud, and Van Reenen 2023). At the core of these propositions lies an unanswered question of whether government regulation increasingly burdens small and large firms differently and acts as an obstacle to firm growth.¹ Due to limited information on firm-level incidence of regulatory compliance costs, prior studies have mostly focused on how industries' overall dynamism relates to industry compliance requirements, or at best how small and large firms' growth may respond to industry-specific regulatory policy changes. While these estimates are informative about small and large firms' elasticity to specific regulations, they cannot assess the full extent of the regulatory burden on small and large firms, which is the central metric for a complete quantitative assessment.

Indeed, how regulatory compliance costs fall on small and large firms is unclear *ex ante*. On one hand, one could posit that the United States' regulatory system may entail proportionally lower compliance costs for large businesses due to fixed costs or lobbying (Davis 2017; Alesina, Battisti, and Zeira 2018; Gutiérrez and Philippon 2019; Akcigit and Ates 2021; Aghion, Bergeaud, and Van Reenen 2023). On the other hand, small businesses may pay lower compliance costs than others due to the many exemptions stemming from regulatory tiering (Brock and Evans 1985; Aghion, Bergeaud, and Van Reenen 2023) or to more relaxed enforcement in the inspection of micro-establishments. Nonetheless, recent studies suggest that substantial aggregate output and productivity losses can arise in the presence of size-dependent regulatory constraints (Restuccia and Rogerson 2008; Guner, Ventura, and Xu 2008; García-Santana and Pijoan-Mas 2011; Hovenkamp and Morton 2019), yet, their extent is unknown.

While these questions are central to the ongoing policy debate on the regulatory compliance burden, measuring the size-dependency of regulatory compliance across the full spectrum of U.S. firms is challenging. For Goff (1996, p. 87) "the measurement problems present such a large barrier that one could flatly assert the total amount of regulation to be unmeasurable by direct observation." Several factors make assessing the incidence of the regulatory burden across firms difficult from the quantitative perspective. First, regulatory requirements are different across

¹ Gutiérrez and Philippon (2019, p. 6) highlight the importance of "emphasizing the heterogeneous impact of regulations on small and large firms" and make a novel empirical contribution towards this direction.

industries, from financial services to mining, and from agriculture to pharma, to name a few. Second, firms are subject to multiple sources of regulation, from federal to state and local agencies, and these overlapping regulations often intersect or complicate each other (Agarwal et al. 2014; Kalmenovitz, Lowry, and Volkova 2025). Third, many regulators provide specific exemptions from compliance and rules are tiered to be less burdensome for certain employment classes, business forms, or ownership structures. Examples include small businesses as defined by the U.S. Small Business Administration,² sole proprietorship versus C-corporations, or publicly traded versus privately held firms. Fourth, for given regulatory requirements, small and large firms may face different stringency or frequency of inspection (Stiglitz 2009; Kang and Silveira 2021), which may result in heterogeneous on-equilibrium compliance efforts. Fifth, small and large firms face different technological constraints and reach different efficiency levels in complying with regulations, for example, due to economies of scale or fixed costs.

A first contribution of this paper—and a necessary step for studying the relationship between regulation and firm size—is to systematically measure the costs of regulatory compliance for establishments and firms. We start by focusing on labor costs for regulatory compliance, which by many accounts are a central component of direct compliance costs.³ Our measure is made possible by merging occupational task data from O*NET (V23.0) with the Occupational Employment and Wage Statistics (OEWS) establishment-occupation level microdata from the U.S. Bureau of Labor Statistics (BLS), a large representative survey covering about 1,200,000 establishments from all industries for 2002–2014.

We start by measuring the regulation-relatedness of a labor task in Section 2. Using a combination of keyword matching, manual assignment, and natural language processing methods, we assess the regulation-relatedness of 19,636 tasks in O*NET. We next aggregate the regulation-relatedness of tasks to the occupation level using an importance scale of each task for each occupation also provided by O*NET. Regulation-related occupations span about one-third of all occupation categories, covering not only compliance officers, but also compensation and benefits managers, financial examiners, nurse practitioners, insurance policy processing clerks, environmental technicians, among many others. We obtain our

² See, for example, the U.S. Small Business Administration Office's advocacy for avoiding "excessive regulatory burdens on small businesses" at <https://advocacy.sba.gov/about/> (Last accessed 5/1/23).

³ For example, Jamie Dimon, CEO of J.P. Morgan Chase, in a "Letter to Shareholders" on April 9, 2014, states that "We are applying enormous resources to the task [of regulatory compliance]... 13,000 employees will have been added since the beginning of 2012 through the end of 2014 to support our regulatory, compliance and control effort." We discuss later systematic evidence on the importance of labor spending for total regulatory compliance costs. As a robustness check, we also complement our measure by accounting for capital costs later.

key measure of regulation intensity for each establishment by aggregating occupations' regulation-relatedness weighted by the establishment's labor spending on each occupation in OEWS. We label this measure *RegIndex*, a variable that indicates the percentage of an establishment's total labor spending ascribed to performing regulation-related tasks. Using the establishment's employer identification number (EIN), we further aggregate *RegIndex* at the firm level (Song et al. 2018), and perform all analyses at both the firm and the establishment level, unless otherwise noted.

Compared to existing approaches for measuring the regulatory burden on industries or publicly traded firms (discussed in Section 1), our measure has several novel and important features. First, it provides a firm-specific micro measure, crucial for capturing heterogeneity in regulatory burden, as the same regulation may affect different firms differently, even within the same sector. Second, our results are based on actual compliance costs paid by firms rather than on statements, expectations, or projections by management or government agencies. Third, our measure reflects on-equilibrium effects of regulation, incorporating firms' endogenous responses to enforcement, regulatory ambiguity, and related factors. Fourth, our analysis covers both very large and very small firms—an essential feature for assessing the impact of government rules on small businesses. Fifth, our approach encompasses regulations from all relevant sources, including federal, state, local, and industry-level privately enforced rules. Finally, our methodology is operational and reproducible by government agencies for ongoing assessments and validation of rules; for instance, it can serve as a straightforward empirical measure of ex post compliance costs, complementing the White House Office of Information and Regulatory Affairs' (OIRA) ex ante projections.

Using a broad definition of regulation-related tasks that includes tax compliance, an average firm spends 3.33% of its total labor costs on performing regulation-related tasks per year; under our most conservative measure, the average is 1.34%. The aggregate compliance hours implied by *RegIndex* correspond to a range between 49 and 121% of the total compliance hours estimated by regulatory agencies from OIRA reports.⁴ When aggregating *RegIndex* to the national level, we observe that regulatory compliance costs of U.S. businesses have grown by about 1% each year from 2002 to 2014 in real terms.

We next validate our *RegIndex* measure through a battery of tests. First, we show that our *RegIndex* can successfully pick up major industry-level

⁴ We use the most conservative version of *RegIndex*, which we have constructed throughout our analyses. All results reported in the paper are robust to using a broad version of *RegIndex* that includes tax compliance and a medium version that excludes taxes, but does not downweight regulation-related tasks based on the presence of other task objectives or keywords. These are available from the authors upon request.

regulations and, importantly, also deregulations, which is a challenging task for many existing measures in the literature. Second, we find that the aggregated RegIndex series, which adds up establishments' actual regulatory compliance costs, closely tracks the aggregate predicted compliance costs reported by the OIRA. Third, we estimate a state-level RegIndex measure and show that states leaning towards voting for the Republican Party tend to have lower RegIndex. Fourth, we construct RegIndex for publicly traded firms before and after IPO and show that RegIndex captures the increased regulatory burden after firms become publicly traded (Ewens, Xiao, and Xu 2024; Coates IV 2007; De Fontenay 2016). Moreover, our RegIndex for publicly traded firms are also positively correlated with existing text-based firm-level measures from Kalmenovitz (2023) and Armstrong, Glaeser, and Hoopes (2025).

After discussing labor costs of regulatory compliance, we extend the analysis by including establishments' capital equipment and tools expenditure related to regulatory compliance. Capital equipment costs are measured following the approach presented in Caunedo, Jaume, and Keller (2023). We use this methodology to create an extended version of regulatory costs, including capital to generalize and assess the robustness of our findings. Indeed, the inclusion of capital costs produces results consistent with what we observe using labor costs alone, but the compliance cost levels are about 20% higher.

The paper then proceeds to address its key question. Section 3 asks how regulatory costs change with respect to firms' (or establishments') size, and it investigates the presence of increasing or decreasing returns to scale in regulatory compliance. Intuitively, a regulatory compliance cost schedule that is non neutral to scale—as approximated by total employment—may distort incentives for producers, induce factor misallocation (Hsieh and Klenow 2009; Huneeus and Kim 2018), and constrain productivity growth (Parente and Prescott 2002; Aghion, Bergeaud, and Van Reenen 2023). On one hand, increasing regulatory costs with size implies incentives for firms to remain small, below the efficient scale of production (Guner, Ventura, and Xu 2008; Garicano, Lelarge, and Van Reenen 2016). This may arise, for instance, from government policies designed to support small businesses through lighter regulation (regulatory tiering), which implies increasing compliance requirements kicking in at progressively higher levels of employment (Brock and Evans 1985; Brock, Evans, and Phillips 1986; Alvero, Ando, and Xiao 2023; Ewens, Xiao, and Xu 2024).⁵ On the other hand, decreasing regulatory costs with scale favors larger players

⁵ The distortive effects of regulatory tiering are well documented, especially in Europe, where firms below efficient scale appear over-represented and mid-sized firms under-represented. See Evans (1986); Boeri and Jimeno (2005); Schivardi and Torrini (2008); Gouriou and Roys (2014); Garicano, Lelarge, and Van Reenen (2016), among many others.

vis-à-vis smaller competitors, quashing entry and fostering concentration (Gutiérrez and Philippon 2019; Philippon 2019). This may arise naturally from economies of scale in regulatory compliance, for example, due to the presence of fixed costs,⁶ or it may even derive from regulatory capture and special deals for large players.⁷ To the best of our knowledge, extant literature offers no clear consensus on the shape of the returns to scale in regulation for the entire U.S. economy.

Our main finding in Section 3 is an inverted-U relation between RegIndex and firm size. In particular, firms under 500 employees experience increasing regulatory compliance costs as a share of total wages, with the percentage of labor spending on regulatory compliance sharply increasing with employment. This is consistent with the Code of Federal Regulation having elements of tiering, but also with smaller firms having lower incentives to comply with regulations given a higher likelihood of flying under the regulator's radar (see Section 4). For firms above 500 employees, economies of scale kick in, and the percentage of labor spending on regulatory compliance progressively decreases with employment. On average, RegIndex for mid-sized firms is about 47% greater than that of the smallest firms and 18% greater than that of the largest firms.

The inverted-U relation between RegIndex and firm size is a particularly robust feature of the data. First, we observe this relation after controlling for industry fixed effects, state fixed effects, and even firm fixed effects. Second, the relation remains non-monotonic after including capital costs in the RegIndex construction and when we probe the regulatory requirements of the same occupation across firms of different sizes. Third, we extensively examine the impact of outsourcing by conducting two industry-specific case studies based on reported size-specific outsourcing intensities and developing a comprehensive estimation that assigns local compliance outsourcing providers' costs to firms in different size bins, and we show that accounting for outsourced compliance costs is unlikely to overturn our finding that small firms have lower RegIndex than mid-sized firms. Fourth, we examine whether small and large firms have different requirements of regulatory compliance for the same occupation by analyzing additional data from 14 million job postings in the Burning Glass Technologies database. We show that for job postings of the same occupation, mid-sized firms require more regulatory compliance skills in their descriptions

⁶ Regulatory environments where increasing returns to scale prevail are frequently documented in the literature. For instance, in the case of environmental regulation, List et al. (2003) examine the effects of air quality regulation on new plant formation, finding large negative impacts in New York State in 1980–90. In the case of pharmaceutical companies and the Food and Drug Administration, Thomas (1990) also finds large negative effects on small firms. In their analysis of compliance risk, Davis (2017) and Calomiris, Mamaysky, and Yang (2020) find that larger corporations are at an advantage (while focusing only on public companies).

⁷ This is relevant with respect to classic research on the political economy of regulatory capture. For a recent discussion, see Lancieri, Posner, and Zingales (2022).

than do small and large firms, further reinforcing the finding of a baseline inverted-U shape between RegIndex and firm size.

Section 4 investigates three mechanisms behind the inverted-U shape relationship between RegIndex and firm size: (i) fixed costs in compliance, (ii) size-dependent regulatory requirements, and (iii) differential enforcement for large and small firms. All together, the mechanisms of fixed costs and regulatory tiering appear consistent with key parts of the non-monotonic relation between compliance costs and firm size, while we find no evidence of a sizable role for differential enforcement. Specifically, in terms of (i) fixed costs, we find that large firms tend to systematically hire more specialists to comply with regulation than mid-sized and small firms. This evidence is consistent with the centralization of regulatory compliance: large firms find it economical to consolidate their compliance efforts in the hands of dedicated employees to save costs.

We propose and implement a shift-share method for understanding whether part of the relation between RegIndex and firm size may be driven by (ii) size-dependent regulatory requirements or by (iii) regulatory agencies' heterogeneous enforcement efforts. Our methodology has two steps. First, we categorize regulation-related tasks into tasks for coping with the enactment and retirement of regulations, that is, requirement-sensitive tasks, and tasks for coping with changes in regulatory enforcement, that is, enforcement-sensitive tasks. To do so, we exploit the fact that each regulation-related task has a different exposure to the 12 major regulatory agencies, that is, the "share," due to the task's presence in different industries and different industries' exposure to the agencies. For each of the major regulatory agencies, we construct a "shift" in its regulatory requirements based on the enactment and retirement of the agency's regulations and a "shift" to its enforcement stringency based on changes in its regulatory employment. Estimating a business's spending on each task in response to the two shift-share measures, we obtain each regulation-related task's sensitivities to regulation requirement and enforcement and its categorization. Second, we use the two types of tasks to decompose a firm's (establishment's) RegIndex into RegIndex for requirement and RegIndex for enforcement. We show that the fact that small businesses are systematically shielded from regulatory requirements (that is, regulatory tiering) is a mechanism matching the upward component of the inverted-U relation between regulatory compliance costs and firm size, but no evidence of differential enforcement effort by government agencies playing a role. This finding is consistent with the pervasive use of threshold-based regulations in the United States.

Finally, we note that the inverted-U pattern can have implications for the dynamics of the U.S. firm size distribution. While our setting is not well suited to establish causality (see [Garicano, Lelarge, and Van Reenen \(2016\)](#) for a proper setting), with this caveat in mind, we document two

descriptive patterns: First, we find a strong negative association between RegIndex and changes in the firm size distribution within sectors. Second, at the aggregate level, we observe a rise in the share of very small firms (fewer than 10 employees) and a corresponding decline in the share of mid-sized firms between 2002 and 2014. We view these patterns as suggestive, and hope they can help guide future work on this topic.

Our paper confines its scope to the costs of regulation without addressing the benefits of regulation. We further limit our analysis to assessing the labor and equipment capital costs of regulatory compliance in terms of workers' wages and associated equipment and tools costs paid to comply with rules or standards required by the government for existing firms. Hence, our work does not cover other types of regulatory compliance costs, such as non-equipment expenditures (e.g., structure capital like reinforced concrete walls or pumping or draining infrastructure for mine water.) and foregone investment opportunities and profits due to regulatory risk.⁸ We also do not separately capture setup fixed costs for compliance as "one-time costs of learning the relevant regulations, developing compliance systems and establishing relationships with regulators," which may prove substantial barriers, as discussed in [Davis \(2017, p. 14\)](#), [Alesina, Battisti, and Zeira \(2018\)](#), [Aghion, Bergeaud, and Van Reenen \(2023\)](#) and systematically documented at least since [Djankov et al. \(2002\)](#). Nonetheless, we argue that labor and equipment compliance costs are a key dimension of the question. According to survey estimates from the [Securities Industry Association \(2006\)](#), 93.9% of regulatory compliance costs in the U.S. financial sector are labor related and 3.3% are physical capital related.⁹ According to survey estimates from the [National Association of Manufacturers \(2014\)](#), 68.4% of regulatory compliance costs in the U.S. manufacturing sector are labor related and 13.4% capital related. Finally, while we do not directly observe firms' compliance costs paid to outsourcing service providers (e.g., external legal, compliance, and accounting services), the aforementioned surveys show that spending on outside advisers accounts for only 2.8% and 8.7% of total compliance costs for the financial sector and the manufacturing sector, respectively. We also conduct extensive assessments, which suggest that accounting for outsourced compliance costs is unlikely to affect our main findings.

⁸ See, for example, [Ryan \(2012\)](#) for a case where such omissions are of the first order. For analysis of the social costs of environmental regulation, see [Hazilla and Kopp \(1990\)](#).

⁹ However, see [Alvero, Ando, and Xiao \(2023\)](#) for a criticism of self reported regulatory costs for the banking sector.

1. Related Literature

This paper relates to several strands of literature. The first pertains to measurement. While we are not aware of any alternative measure of regulatory costs with the exact properties of RegIndex, it is important to mention that other methods designed to obtain a comparative perspective of regulatory requirements have been proposed in the past, at least since [Morrall \(1986\)](#), and that many have important advantages of their own.¹⁰ [Al-Ubaydli and McLaughlin \(2017\)](#)'s RegData, a data repository maintained by the Mercatus Center at George Mason University, is built using QuantGov, a library of machine learning algorithms and text analysis tools designed to collect information about the number of restrictions, rule complexity, and industry incidence from the text in the Code of Federal Regulations. In another novel application, [Kalmenovitz \(2023\)](#) employs data from Form 83-I filed with OIRA to indicate the expectations of regulators about the cost of compliance with each regulation.¹¹ These measures shed light on regulatory burden at the more aggregate level.

More recently, a burgeoning strand of literature has developed to measure publicly traded firms' regulation. [Davis \(2017\)](#) focuses on Part 1A of 10-K filings to gauge publicly traded firms' exposure to regulatory and policy risk.¹² [Calomiris, Mamaysky, and Yang \(2020\)](#) focus on the transcripts of corporate earnings calls made by publicly traded firms and show that its measure of regulation is predictive of sales and assets growth, leverage, and other metrics of firm performance.¹³ [Kalmenovitz \(2023\)](#) uses supervised machine learning to assign paperwork regulations (Form 83-I) to each publicly traded firm based on the similarity between the firm's 10-K (Item 1) text and the regulation's text. A more recent development by [Armstrong, Glaeser, and Hoopes \(2025\)](#) measures each publicly traded firm's exposure to each government regulatory agency based on 10-K texts,

¹⁰ A few earlier studies attempted to use occupational data to measure the effects of regulation as we do. For example, [Schaefer and Zimmer \(2003\)](#) use accountants and auditors from the Current Population Surveys to examine the state regulations in the accounting profession from 1984 to 2000. [Parker, Rosen, and Nielsen \(2009\)](#) survey 999 large Australian firms regarding their expenses on lawyers and compliance professionals to measure firms' costs of compliance with the Trade Practices Act.

¹¹ The 83-I forms include estimates of how many responses the regulator will receive per year; how many work hours firms will be required to dedicate to complying with the regulation; and, for 20% of the regulations, the estimated dollar costs of compliance ([Kalmenovitz 2023](#)). Relative to [Al-Ubaydli and McLaughlin \(2017\)](#), the method in [Kalmenovitz \(2023\)](#) has the major benefit of aggregating the expectations of experts regarding the compliance burden.

¹² This is also an approach preceded by [Baker, Bloom, and Davis \(2016\)](#) and followed subsequently by [Hassan et al. \(2019\)](#), who focus on firms' exposure to a broader political risk, as expressed in the 10K-forms of publicly traded firms.

¹³ An important advantage of the [Davis \(2017\)](#) and [Calomiris, Mamaysky, and Yang \(2020\)](#) approaches is that they are apt at capturing future regulatory risk, both in terms of discretionary enforcement and of new rules affecting firms. Indeed, [Calomiris, Mamaysky, and Yang \(2020\)](#) underscore compliance risk as the most relevant channel behind their finding, rather than physical compliance costs.

which further allows researchers and policy makers to assess the impact of regulation from each agency on firms.¹⁴

Our RegIndex measure complements these existing firm-level measures in at least two unique ways. First, our measure can be applied to both publicly traded firms and also privately held firms. Hence, it can help us to understand regulatory costs across the full distribution of firms, including the very small firms with fewer than 50 employees that are frequently in the policy debate. Our measure can also help us to understand how publicly traded firms' regulatory burden changes before and after an IPO, as we demonstrate in this study. Second, our measure provides a de facto dollar amount of each firm's regulatory compliance costs, which is difficult to obtain with previous measures based on firms' 10-K or earnings call texts. Our measure and data also face limitations, as accessing our firm-level RegIndex measure requires the approval of the BLS for the use of their confidential microdata. BLS surveys most establishments once every 3 years, which further limits future research on within-establishment changes in RegIndex to a 3-year horizon rather than higher frequencies. To shed light on how our measure compares with existing measures, we compute RegIndex for publicly traded firms and show that our RegIndex is about 17%–28% correlated with the measures from [Kalmenovitz \(2023\)](#) and [Armstrong, Glaeser, and Hoopes \(2025\)](#), suggesting that our measure captures related but also distinct dimensions of regulatory burden.¹⁵ We refer to [Bombardini, Trebbi, and Zhang \(2025\)](#) for more detailed discussions on recent methodologies on measuring costs and benefits of regulation and exploring their economic impacts on firms.

The second strand of literature we contribute to is on the distribution of regulatory burden across firm sizes. A large body of prior work has examined this question in specific settings and showed inconclusive findings regarding which group is more burdened by regulations. On one hand, larger firms may bear a lower percentage of the regulatory burden because of economies of scale or regulatory capture. Prior studies supporting this view usually draw conclusions by examining small and large firms' economic responses to regulatory changes instead of directly measuring the heterogeneous regulatory costs by firm size. For instance, [List et al. \(2003\)](#) and [Thomas \(1990\)](#) show that specific regulations more negatively affect smaller firms, [Gutiérrez and Philippon \(2019\)](#) and [Philippon \(2019\)](#) more systemically show that the employment of smaller firms is more negatively associated with industry regulation using RegData, and [Davis \(2017\)](#) and [Calomiris, Mamaysky, and Yang \(2020\)](#)

¹⁴ See also [Bombardini, Trebbi, and Zhang \(2025\)](#) for a recent review of the economics literature on measuring the costs and benefits of regulation.

¹⁵ We thank Joseph Kalmenovitz, Daphne Armstrong, Stephen Glaeser, and Jeffrey Hoopes for making their firm-level measures publicly available.

find that larger corporations are at a greater advantage in responding to compliance risk than while focusing on publicly traded firms. Among studies supporting this view, they also make different predictions as to how regulation shapes firm size distribution: [Klapper, Laeven, and Rajan \(2006\)](#) show that regulation forces successful entry firms to be larger, whereas [Singla \(2023\)](#) shows that increases in regulatory costs contribute to small firms getting smaller. Reconciling this debate is challenging as it is difficult to directly observe regulatory costs for small firms. On the other hand, small firms may bear a lower regulatory burden due to regulatory tiering or lax inspection by regulatory agencies. Prior studies supporting this view usually draw conclusions by examining firm size distribution above and below the tiers rather than directly showing firms' compliance costs ([Brock and Evans 1985](#); [Brock, Evans, and Phillips 1986](#); [Alvero, Ando, and Xiao 2023](#); [Ewens, Xiao, and Xu 2024](#)).

Our study contributes to this literature by directly showing regulatory compliance costs across the full distribution of firm sizes, with some novel implications. We show a more nuanced picture relative to the extant literature by highlighting a non-monotonic relationship between regulatory burden and firm size. Our finding that mid-sized firms are more burdened than small and large firms offers a potential reconciliation of prior studies, which tend to focus on firms from the two ends of the size distribution. Our finding also has important implications for the equilibrium size distribution of firms, as it indicates that the heaviest regulatory hurdle for firm growth is when firms become mid-sized. We further analyze the mechanisms behind these results by highlighting the differential roles of regulatory requirements versus endogenous compliance to enforcement.

Finally, this paper is related to a newly revived literature on regulation and its political economy. Part of this literature focuses on the role of firms in influencing regulation, via lobbying and political influence ([Blanes i Vidal, Draca, and Fons-Rosen 2012](#); [Bombardini and Trebbi 2025](#); [Bertrand, Bombardini, and Trebbi 2014](#); [Kang 2016](#); [Bombardini and Trebbi 2020](#)) or, more recently, via direct contest between different groups of firms ([Singla 2023](#)). Another branch explores the political and policy risks to which firms are exposed ([Julio and Yook 2012](#); [Baker et al. 2014](#); [Baker, Bloom, and Davis 2016](#); [Hassan et al. 2019](#)). A burgeoning recent literature on the interaction between market power and political power centers on regulation as the main tool employed by large business entities to generate rents ([Callander, Foarta, and Sugaya 2021](#); [Cowgill, Prat, and Valletti 2021](#)).

2. Data, Measure, and Validation

2.1 Data

Our main source of information is the establishment-occupation-level microdata from 2002 to 2014 provided by the OEWS program of the BLS.

This data set covers surveys that track employment and wage rates for over 800 detailed occupations in approximately 1.2 million establishments over the course of a 3-year cycle (Zhang 2019). The sample of establishments covers, on average, 62% of non-farm employment in the United States. Within a 3-year cycle, 400,000 establishments are surveyed during each year and therefore the same establishment is surveyed at most every 3 years (e.g., in t and $t + 3$). The microdata provides each establishment's sampling weight (to recover economy-wide aggregates), NAICS 6-digit industry code, county code, government ownership indicators, and the parent firm's employer identification number.

Defining the appropriate regulatory compliance entity is nontrivial. A firm may be considered the ultimate entity, in that it pays the fines and penalties if inspected at any constituting establishment and found in violation of agency rules. However, regulation varies across industries and states, inducing a firm to spend different amounts of resources on regulatory compliance across different constituting establishments. For these reasons, we conduct our main analysis at both the firm level and the establishment level.

An EIN will define the boundary of a firm in our analysis. This is because the EIN identifies a firm for tax purposes, and also because the EIN is commonly used to define a firm both in the academic literature (e.g., Chodorow-Reich 2014; Song et al. 2018) and by government agencies, such as the IRS and the BLS.¹⁶ Following the convention recommended by the OEWS program, we aggregate establishments of an EIN surveyed in years $t - 2$ to t to represent the occupational composition of the EIN in year t .¹⁷ If a firm has establishments spanning multiple industries, we define the firm's major industry as the NAICS 6-digit code with the highest employment share.¹⁸

Our research makes use of the task statements for each occupation from the O*NET (V23.0) database. Each task statement is a single sentence and is pertinent to a unique occupation. An occupation is described by an average of 22 different task statements. O*NET also provides an average rating indicating how important the task is for the occupation rated by incumbent workers working in the occupation, occupational experts, and analysts. Occupations are categorized at the 8-digit standard occupational code (SOC) level. We match the 8-digit SOC in O*NET to the 6-digit SOC

¹⁶ See U.S. Department of Labor, Bureau of Labor Statistics, "Business Employment Dynamics Size Class Data: Questions and Answers," <http://www.bls.gov/bdm/sizeclassqanda.htm>, questions 3 and 5.

¹⁷ The BLS OEWS program uses a similar aggregation approach to publicize industry- or geographical-level statistics for each year. See technical notes at https://www.bls.gov/oes/oes_4oc_4rch.htm.

¹⁸ Because the OEWS survey does not cover all establishments of a firm, our firm-level measures contain measure errors and our firm-level employment is underestimated. Given that over 80% of establishments in our sample are standalone single-unit firms (Ayyagari and Maksimovic 2017), the establishment-level and firm-level results are qualitatively the same for most analyses, as we will show.

occupations in the OEWS microdata, creating a characterization of all tasks performed at the establishment level, which allows for the construction of the measures described in the following subsections.

2.2 A simple theoretical framework for the measure

Our approach to measuring regulatory compliance costs for firms (or establishments) is as follows. Let us assume that a firm's labor includes workers performing regulatory compliance-related tasks and workers performing other production tasks. Let L_{it} be the total number of production workers employed in firm i at time t and w_{it} the average wage paid to the workers. Define R_{it} as the total number of workers occupied in regulatory compliance-related tasks. We allow the average wage of compliance workers to differ from that of production workers (due to specialization) and indicate it with w_{it}^r . We assume all wages to be taken as given from the firm's perspective. R_{it} can be derived as the result of optimization on the part of the firm owners, who maximize profits:

$$Y_{it} - w_{it}L_{it} - w_{it}^r R_{it} - p_{it} \times f_{it}$$

with respect to R_{it} and L_{it} , where firm i faces a probability of inspection p_{it} at period t and fines $f_{it} = \frac{\tilde{R}_{it}}{R_{it}}$ are levied in case the firm is found in less than full compliance with regulatory requirements $\tilde{R}_{it}$. We further assume a constant return to scale production function $Y_{it} = \phi_{it}L_{it}$, where ϕ_{it} indicates a firm-specific productivity shock, and that $\tilde{R}_{it} = \tilde{R}(L_{it}) = \rho L_{it}^\alpha$ and $p_{it} = \pi L_{it}^\beta$, where α, β, ρ, π are policy parameters governing regulatory requirements and enforcement effort targeting establishment i . Simple static profit maximization implies

$$R_{it}^* = \left(\frac{p_{it} \tilde{R}_{it}}{w_{it}^r} \right)^{\frac{1}{2}} = \left(\frac{\phi_{it} - w_{it}}{(\alpha + \beta)(w_{it}^r \pi \rho)^{\frac{1}{2}}} \right)^{\frac{\alpha + \beta}{\alpha + \beta - 2}} \left(\frac{\pi \rho}{w_{it}^r} \right)^{\frac{1}{2}}$$

$$L_{it}^* = \left(\frac{\phi_{it} - w_{it}}{(\alpha + \beta)(w_{it}^r \pi \rho)^{\frac{1}{2}}} \right)^{\frac{2}{\alpha + \beta - 2}}. \quad (1)$$

Note that, intuitively, equilibrium compliance is an increasing function of regulatory requirements $\tilde{R}_{it}$; an increasing function of the enforcement activity of regulators via inspections, p_{it} ; and a decreasing function of regulatory compliance labor wages, w_{it}^r .

We define the index of regulatory compliance costs as:

$$\text{RegIndex}_{it} = \frac{w_{it}^r R_{it}}{W_{it}}, \quad (2)$$

that is, the share of the total wage bill $W_{it} = w_{it}^r R_{it} + w_{it} L_{it}$ allocated to compliance labor related to regulation.

2.3 Construction of the regulation index

The construction of the regulatory index, RegIndex, starts with identifying which tasks are related to regulatory compliance. We achieve this goal by analyzing the texts of task statements in the O*NET data in two phases: a keyword matching phase and an annotation phase. In what follows, we briefly describe the procedure for estimating our preferred (conservative) version of the RegIndex, which we use for our analysis, and we relegate more details and further description of two broader versions of the RegIndex to the [Internet Appendix A](#).

In the keyword matching phase, we identify regulation-related tasks by matching the task statements to two tiers of keywords to balance the rate of false positives and false negatives in identifying regulation-related tasks. The first tier of keywords includes the words “regulation,” “regulations,” and “regulatory.” These matches exhibit a low rate of false positives, but may miss some regulation-related tasks when regulation is described using synonyms. For this reason, a second tier of keywords includes 34 keywords that identify alternative references to regulation, such as “statutes,” “ordinances,” and “code,” references to government agencies, and actions of compliance. [Internet Appendix A](#) details the 34 keywords, and Table IA.1 lists 10 examples of regulation-related tasks identified by the first-tier keywords.

In the annotation phase, we manually annotate each matched task statement to exclude further false positives, such as tasks in which the word “codes” refers to computer programming. This procedure results in a final list of 829 regulation-related tasks out of a total of 19,636 tasks in the O*NET database. Next, we assign a regulation-relatedness value between 0 and 1 to account for the heterogeneity among regulation-related tasks. In particular, tasks identified by the second-tier keywords, which may be less informative about the tasks’ relevance for government regulation, and tasks with multiple focuses in their statements are downweighted to have a regulation-relatedness value less than 1.

Having measured the regulation-relatedness of each task, we next compute the regulation-task intensity (RTI) for each SOC 6-digit occupation by averaging its tasks’ regulation-relatedness values weighted by the importance ratings of each task for that occupation based on O*NET importance weights. Table IA.2 lists the top 25 SOC 6-digit occupations with the highest RTI. For instance, Compliance Officers have the highest RTI of 0.343, meaning that, conservatively, Compliance Officers on average spend 34.3% of their work hours directly performing government regulation-related tasks (Compliance Officers also perform tasks such as maintain and repair materials, worksites, and equipment, which we do not

classify as regulation). Finally, we merge each SOC 6-digit occupation's RTI to the relevant establishments in the OEWS data, which provides each establishment's labor cost (employment times wage rate) for each occupation.

We define an establishment's regulation index (RegIndex) as the share of its total labor cost spent on performing regulation-related tasks. In particular, an establishment i 's RegIndex is its occupations' average RTI weighted by its labor cost on each occupation j at time t :

$$\text{RegIndex}_{i,t} = \frac{\sum_j \text{RTI}_j \times \text{emp}_{i,j,t} \times \text{wage}_{i,j,t}}{\sum_j \text{emp}_{i,j,t} \times \text{wage}_{i,j,t}} \times 100. \quad (3)$$

Using regulation-relatedness values from the conservative, medium, and broad approaches, we obtain three versions of establishment-level RegIndex. The three versions of RegIndex are highly correlated. Hence, we perform much of our analysis using the conservative RegIndex.¹⁹

A firm's RegIndex is similarly obtained based on occupational labor costs aggregated from those of establishments with the same EIN. Table IA.3 shows the top 25 NAICS 3-digit industries whose establishments have the highest average RegIndex. This study focuses on firms' (or establishments') in-house regulatory compliance cost. Hence, it excludes firms in industries that provide legal or compliance work as their main function, including legal services, accounting services, central banks, and public administration.²⁰

Capital. Our baseline RegIndex is based on the firms' labor costs. As noted earlier, another important component of firms' regulatory compliance spending is capital expenditure. Unfortunately, due to the lack of microdata on firm-level detailed capital spending, a precise estimation of firms' regulatory compliance costs accounting for capital expenditure is not available. Nevertheless, alternatives have been devised in the literature based on tools and capital equipment use. Here we follow [Caunedo, Jaume, and Keller \(2023\)](#) and construct an approximate measure of capital cost per employee for each occupation, based on the occupation's use of tools and equipment from the O*NET Tools and Technology module and from the National Income and Product Accounts (NIPA) quantity and price tables for each equipment item. With this additional information available, one is able to sum up capital user costs per employee and annual wages per employee for each occupation, obtaining the total input cost (in terms of

¹⁹ Notice that the definition of RegIndex can also be applied to a single regulation related task o by assessing the ratio of the task's associated labor costs as a share of the total wage bill of the firm or establishment. We indicate it as $\text{RegIndex}_{i,o,t}$.

²⁰ Following [Song et al. \(2018\)](#), educational institutions are also excluded due to high government ownership.

labor and capital) for each employee in each occupation. From this, it is possible to proceed to construct a more general version of RegIndex.

The capital augmented $\text{RegIndex}_{i,t}^{\text{tot}}$ is the share of a firm i 's total input costs at t spent on performing regulation-related tasks:

$$\text{RegIndex}_{i,t}^{\text{tot}} = \frac{\sum_j RTI_j \times \text{emp}_{i,j,t} \times (\text{wage}_{i,j,t} + \text{kcost}_{i,j,t})}{\sum_j \text{emp}_{i,j,t} \times (\text{wage}_{i,j,t} + \text{kcost}_{i,j,t})} \times 100. \quad (4)$$

Focusing on $\text{RegIndex}_{i,t}^{\text{tot}}$, we can derive additional statistics about the the composition of firms' regulatory compliance spending in terms of labor and capital that are highly consistent with industry survey findings.²¹

2.4 Summary statistics

The sample for the analysis in this paper includes 3.03 million U.S. firm-year observations and 3.36 million establishment-year observations surveyed by the OEWS program from 2002 to 2014. Table 1 provides the key summary statistics. To begin with, the unweighted average firm in our sample has 92 employees and the unweighted average establishment has 48 employees; both appear above the national average based on Census statistics. If one applies the sampling weights to the establishments assigned by the OEWS, the weighted average employment falls to 14, which is closer to the establishment-level average employment of 15.6 employees reported by the Census Statistics of U.S. Businesses (SUSB) during 2002–2014. The average annual wage per employee in our sample is \$43,733 (\$2.09 million divided by 47.79), which is in line with the average annual wage per employee of \$41,974 from the SUSB.

The key summary statistics in Table 1 focus on RegIndex. The average firm in our sample spends 1.34% of its total labor costs on regulation-related tasks in any given year. The average establishment spends 1.31%.²² To be further noted, RegIndex varies substantially across firms and establishments, with the 0.5 percentile at 0, the median at 0.8–0.9%, and the 99.5 percentile above 10%. Further decomposition shows that year fixed effects explain merely 0.01% of the variation in establishments'

²¹ For instance, National Association of Manufacturers (2014) shows that labor spending on regulatory compliance cost as a fraction of the sum of labor and capital spending on regulatory compliance cost is 84% for manufacturing industries in 2012 (see Chart 15 of the report). Aggregating regulatory compliance costs based on $\text{RegIndex}_{i,t}^{\text{tot}}$ for the manufacturing sector in 2012, we obtain a corresponding fraction of 78%. Similarly, Securities Industry Association (2006) report shows that labor spending on regulatory compliance cost as a fraction of the sum of labor and capital spending on regulatory compliance cost is 97% for financial industries in 2006 (see Figure 3b of the report). Using $\text{RegIndex}_{i,t}^{\text{tot}}$ for the financial industries in 2006, we obtain a fraction of 88%.

²² These statistics are based on the conservative RegIndex, our main measure. Using the broad measure, the average establishment in our sample spends 3.28% of its total labor costs on regulation-related tasks in any given year, and the average firm spends 3.33%.

Table 1
Summary statistics

Variable	Mean	SD	P0.5	Median	P99.5	Obs.
<i>A: Firms</i>						
<i>Employment</i>	92.16	617.16	1.00	13.00	2,465.00	3,027,680
<i>Annual Wage (\$mn)</i>	4.07	31.30	0.02	0.46	115.48	3,027,680
<i>RegIndex</i>	1.34	1.88	00.86	10.46	3,027,680	
<i>B: Establishments</i>						
<i>Employment</i>	47.79	192.45	1.00	13.00	875.00	3,364,336
<i>Annual Wage (\$mn)</i>	2.09	11.73	0.02	0.44	43.31	3,364,336
<i>RegIndex</i>	1.31	1.90	0	0.80	10.57	3,364,336
<i>C: Industry</i>						
<i>Employment (1,000)</i>	90.66	285.44	0.01	25.13	2,041.20	15,159
<i>Annual Wage (\$mn)</i>	3,611.91	11,112.21	0.12	1,001.32	67,466.05	15,159
<i>RegIndex</i>	1.66	1.02	0	1.60	5.58	15,159

Firms are defined by EIN, following Song et al. (2018). We bundle establishments of an EIN surveyed in year $t-2$ to t as a firm in year t following BLS convention. Industries are defined at the NAICS 6-digit level. We aggregate establishments of an industry surveyed in year $t-2$ to t weighted by the establishments' sampling weights to compute industry-level metrics in t . RegIndex is the ratio of labor spending on regulation-related tasks and total labor spending as a percentage. Sample period is 2002–2014.

RegIndex, state fixed effects explain 0.10%, NAICS 6-digit industry fixed effects explain 36.13% (for coarser categories, NAICS 2-digit, 3-digit, and 4-digit fixed effects explain 8.69%, 12.16%, and 22.15%, respectively), and the residual 63.63% of the RegIndex variation is unexplained by the above.

To illustrate the recent time trends in regulatory compliance costs, Figure 6 plots the aggregate time-series of RegIndex. Following the aggregation method explicit in the OEWS program, Figure 1 aggregates the RegIndex of all establishments in our sample surveyed between $t-2$ and t , weighted by a product of the establishments' sampling weights and their total labor cost, to obtain the average RegIndex of the U.S. economy at t .

From Figure 1, we observe an increase in aggregate RegIndex from 1.49% in 2002 to 1.59% in 2014, reaching \$79 billion in regulatory compliance costs. This level of regulatory compliance costs is a conservative estimate. If we use all private establishments (including establishments that earn revenue from regulatory compliance services, such as legal services), the aggregate regulatory compliance costs are \$103 billion in 2014. If we apply our broad version of RegIndex to all private establishments, the aggregate regulatory compliance costs are \$239 billion. Finally, adding capital equipment brings this upper bound to \$289 billion in 2014. For comparison, U.S. gross business income taxes amounted to \$353 billion in the same year.²³

In real terms, from 2002 to 2014 the yearly growth rate of aggregate regulatory compliance costs averaged around 1%, which is about half of the 1.92% average yearly growth rate of the U.S. Real Gross Domestic Product

²³ See <https://www.irs.gov/statistics/soi-tax-stats-irs-data-book>.

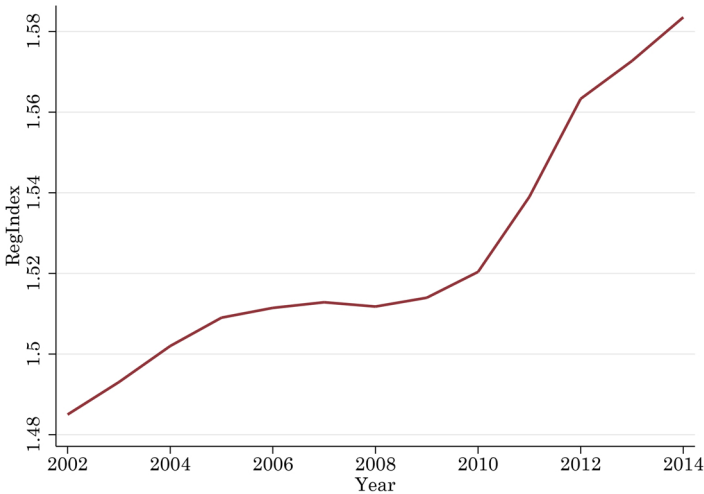


Figure 1
Aggregate series of regulation index

This figure plots the aggregate regulation index from 2002 to 2014. RegIndex is the percentage of an establishment's annual labor spending on performing regulation-related tasks (see Section 2). The RegIndex in each year is aggregated from our final sample of establishments, weighted by establishment weights designated by the BLS. Our final sample focuses on nongovernment/noneducation establishments (Song et al. 2018) and excludes establishments from industries that provide legal or compliance services: legal and accounting firms, government administration, courts, and central banking.

over the same period. Overall, these patterns indicate the substantial economic magnitude of regulatory compliance, but a burden that has been growing at a slower rate relative to the rate of the growth of the U.S. economy.

2.5 Validation

2.5.1 Time-series relation with agency-estimated compliance hours. As a first validation exercise, we examine whether our approach for constructing RegIndex captures the time-series variation in the regulatory costs in the United States. Federal regulatory agencies are required to file Form 83-I to the OIRA in which the agency estimates firms' or individuals' compliance time for each regulation. We collect the estimates for each year from 2002 to 2014 from the "Information Collection Budget of the United States Government" from the White House website.²⁴

We next compute counterpart compliance hours based on our approach for constructing RegIndex. Specifically, we regard an occupation's

²⁴ See <https://www.whitehouse.gov/omb/information-regulatory-affairs/reports/>. The filling for Form 83-I is mandated by the Paperwork Reduction Act (44 U.S.C. 3501). Federal agencies estimate three burden metrics for each regulatory regarding how many responses it will receive per year, how many hours it will take the public to comply with the regulation, and what would be the dollar costs of compliance. Only estimated compliance time is consistently reported by the "Information Collection Budget" report by OIRA each year.

regulation-task intensity as the fraction of time an employee spends on regulation-related tasks in an hour. Assuming that all regulation-related occupations work 2,080 full-time hours in a year (noting that part-time workers are concentrated in the retail and restaurant industries), we estimate U.S. establishments' aggregate de facto regulatory compliance hours.

Figure 2 plots the time-series of annual aggregate regulatory compliance hours based on our approach and the compliance hours estimated by regulatory agencies from OIRA. OIRA compliance hours include a nontrivial amount of estimated hours for households to comply with individual income taxes. To produce a proper comparison of OIRA with our measure, which focuses on businesses' regulatory compliance hours, we exclude estimated hours for individual income taxes from the OIRA total compliance hours.²⁵ Panel A shows that our estimates based on the conservative RegIndex account for about half of the hours estimated by federal agencies through OIRA. Using a less conservative weighting of tasks and accounting for all tax compliance efforts, the compliance hours based on our medium and broad versions of RegIndex amount to 106% and 121% of the OIRA compliance hours respectively.

Figure 2 also shows that compliance hours based on our RegIndex track the estimated compliance hours by agencies robustly over time in terms of changes. The two time-series exhibit a statistically significant correlation of 67% using our main conservative definition and 71% using our medium and broad definitions. Figure IA.1 shows that the results are similar for all three versions of RegIndex.

2.5.2 Establishment RegIndex and industry regulatory policy changes. As a second validation exercise, this subsection illustrates the response of RegIndex to four major industry-level regulatory policy changes. Before and after salient regulatory policy changes, we examine the RegIndex response by establishments within a treated industry relative to appropriately matched control industries. These case studies not only help demonstrate the effectiveness of our RegIndex in tracing major industry-level regulatory policy changes, but also suggest that our RegIndex is able to overcome several limitations of existing regulation measures.

²⁵ Compliance hours for individual income taxes account for about 40% of the Department of Treasury's total regulatory compliance hours. For example, in 2011, The U.S. individual income tax return imposed an estimated 2.70 billion compliance hours out of 6.74 billion compliance hours for Department of Treasury's total regulations, see https://www.whitehouse.gov/wp-content/uploads/legacy_drapal_files/omb/inforeg/inforeg/icb/icb2013.pdf. We thus exclude compliance hours for individual income taxes in Figure 2 by subtracting 40% of the Department of Treasury's estimated compliance hours from the total estimated compliance hours by all agencies. See the breakdown of IRS tax filing hours at <https://taxfoundation.org/tax-compliance-costs-irs-regulations/>.

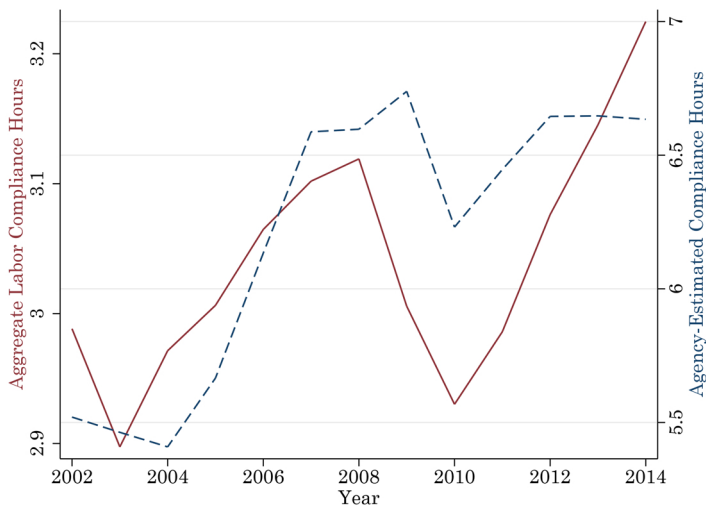


Figure 2
Validation of regulation index using agency-estimated compliance hours

This figure plots the aggregate annual compliance hours (in billions of hours) identified by our most conservative RegIndex measure, which is our baseline measure used in this study. See [Section 2](#) for details about the three versions of RegIndex. We compare the aggregate compliance hours identified by our measure with the estimated annual compliance hours (in billions of hours) submitted by various regulatory agencies to the OIRA. [Figure 1A.1](#) plots additional graphs using the hours identified by two broader versions of the RegIndex measure: Medium RegIndex and Broad RegIndex.

Case 1: Regulation of the Credit Card Industry. The Credit Card Accountability Responsibility and Disclosure Act of 2009 (CARD Act) was drafted to “implement needed reforms and help protect consumers by prohibiting various unfair, misleading and deceptive practices in the [U.S.] credit card market” (U.S. Senate, 2009).²⁶ A key aspect of the CARD Act, in particular, was to impose regulatory limits on the ability of credit card issuers to charge customers certain types of credit card fees; it became effective in February and August of 2010. Accordingly, as the treatment group for this analysis we use establishments in the credit card issuing industry (NAICS 52221). As the control group, we use establishments in all other nondepository credit intermediation industries narrowly defined (NAICS 5222x, except for 52221, including sales financing, consumer lending, and real estate credit).

Case 1 in [Figure 3](#) plots the weighted average RegIndex for the treated credit card-issuing industry and for the control industries in the years around the policy change (from 2005 to 2014). In the figure, establishment RegIndex is weighted by the product of each establishment’s sampling

²⁶ Congress enacted the CARD Act of 2009 after research (e.g., [Simkovic 2009](#)), as cited by a Senate committee, suggests that credit card pricing was less than perfectly competitive.

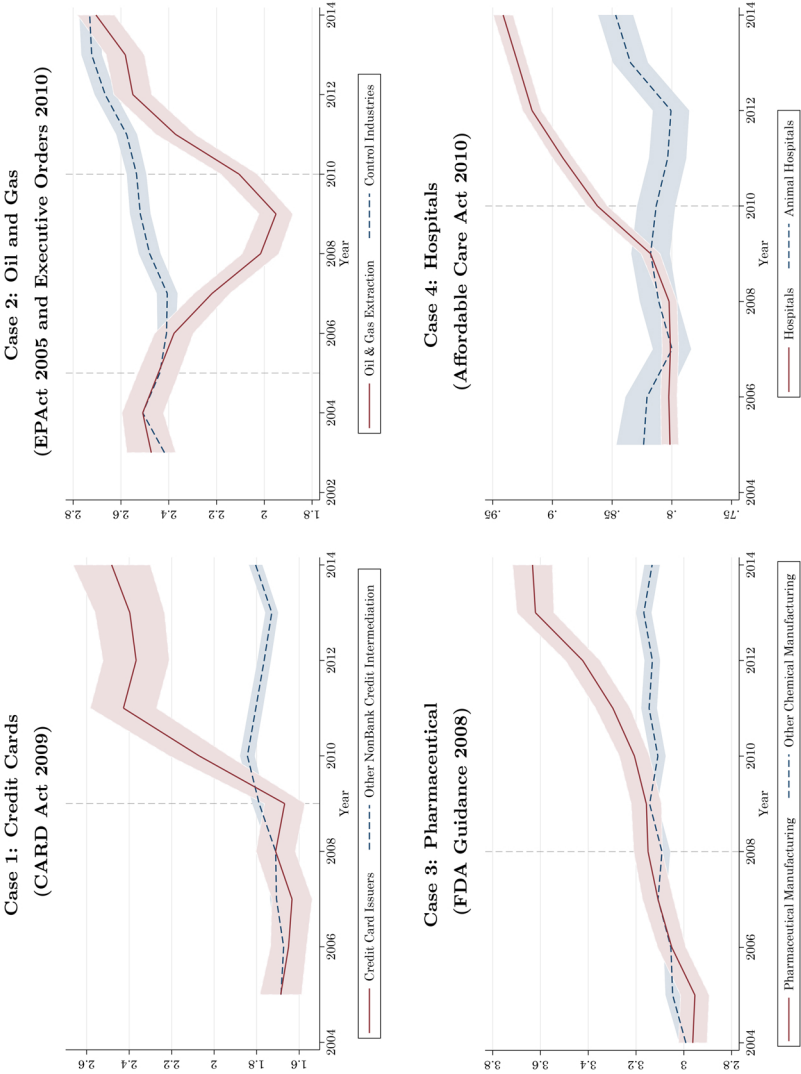


Figure 3

Validation of regulation index using industry regulatory policy changes

This figure plots the response of industries' regulation index (RegIndex) to five industry-level regulatory policy changes. RegIndex is the percentage of an industry's annual labor spending on performing regulation-related tasks. Section 2 provides details of the industry regulatory policy changes and discusses the classification of treated and control groups. To ease the comparison, we shift the lines vertically so that they have the same value in the year before the treatment. The value in the year before the treatment is the average of the regulation measures across the treated and control industries in that year. The difference between the two lines after the treatment, minus the difference between the two lines before the treatment reflects the difference-in-differences estimation. The shaded areas indicate the 95% confidence interval of the industries' average RegIndex.

weight and each establishment's total labor costs. We can clearly observe in this figure that before the enactment of the CARD Act in 2009, the 95% confidence interval of RegIndex for the treated and the control groups appear to overlap and that they are statistically indistinguishable from one another. This feature of the data suggests the validity of the parallel trends assumption necessary for the consistency of the simple difference-in-differences estimator underlying this case study. We can see further that after the policy change, RegIndex in the credit card issuing industry rises dramatically, while the RegIndex average for the control group remains basically flat. While this graphical evidence underscores the ability of RegIndex to trace the heightened regulatory burden associated with the CARD Act on credit card issuing establishments during the post period of the analysis, measuring regulation intensity at such a granular level is proven to be challenging for supply-side approaches. Indeed, we show in Figure IA.2 that the popular RegData measure, which counts restrictive words in the Code of Federal Regulations (CFR), cannot identify a similar effect to ours.

Case 2: *Deregulation and Re-regulation of the Oil and Gas Industry.*

In our second case study, we explore both (i) the deregulation of the oil and gas extraction industry by the Energy Policy Act of 2005 (EPAAct) under President George W. Bush, which exempted oil and gas facilities from environmental regulations to boost production, as well as (ii) the re-regulation of the industry by President Barack Obama's executive orders following the British Petroleum Deepwater Horizon oil spill in 2010. These orders led to several new regulatory policies, which were regarded as "the most aggressive and comprehensive reforms to offshore oil and gas regulation and oversight in U.S. history," according to the Bureau of Ocean Energy Management.²⁷

Our treated industry for this case study is oil and gas extraction (NAICS 2111). We select downstream industries that use a significant amount of the treated industry's output as their inputs as the control industries. We assume that downstream industries share close economic ties with the upstream treated industry, but face a sufficiently different set of regulations to be not as strongly affected by EPAAct and the Obama orders.

In fact, both control and treated industries may be affected by similar economic and regulatory forces prior to the regulatory policy changes.²⁸ Using the U.S. Bureau of Economic Analysis (BEA) input-output table

²⁷ See <https://www.boem.gov/regulatory-reform/>.

²⁸ While intuitive, choosing control industries based on input-output relations offers no guarantee that the treated and control industries will exhibit parallel trends in the regulation index during the pre-treatment periods. We thus examine the parallel trends empirically when analyzing each regulatory shock. Another challenge with this approach is that the control group may also be affected by the new regulatory policy changes. When treatment and control groups are both affected by the regulatory shock, we will be less

from 2007, we select the following three industries as the control group: petroleum and coal products manufacturing (NAICS 3241), natural gas distribution (NAICS 2212), and basic chemical manufacturing (NAICS 3251).²⁹

Case 2 in Figure 3 shows parallel trends of RegIndex for the oil and gas extraction industry and the control industries before the enactment of the EPOA in 2005. After 2005, there appears to be a dramatic decline in the RegIndex for oil and gas extraction relative to the control industries, in line with the EPOA being a deregulatory policy change. This evidence highlights an important advantage of our measure: RegIndex can distinguish between regulation and deregulation. By contrast, separating statutory text that increases regulatory burdens from statutory text that deregulates an industry is challenging for supply-side language-based measures. Such measures tend to increase with both regulation and deregulation. Consistent with this limitation, we show in the Figure IA.2 that the supply-side RegData measure exhibits an increase in regulation of oil and gas extraction after the EPOA was signed into law in 2005.

After the BP oil spill in 2010, consistent with the contemporary understanding that the industry faced heavy re-regulation, we observe a rapid increase in the RegIndex for the oil and gas extraction industry both in absolute terms and relative to control industries.

Case 3: Regulation of Pharmaceutical Manufacturing. Our third case study focuses on pharmaceutical manufacturing which has experienced significant regulatory changes since 2008. The U.S. Food and Drug Administration Amendments Act signed into effect on September 27, 2007 increased the FDA's regulatory authority over drug safety and required companies to implement a risk evaluation and mitigation strategy for many drugs. Following the legislative change, the FDA significantly amended the current good manufacturing practice regulations for human drugs in 2008 on quality control, risk management, record-keeping, facility design and maintenance, and employee training.³⁰

We choose establishments from the pharmaceutical and medicine manufacturing industry (NAICS 3254) as the treatment group for this analysis. As the control group, we use establishments in all other chemical manufacturing industries (NAICS 325x except for 3254) including the manufacturing of basic chemicals, synthetic rubber, pesticides, paints, and

likely to detect significant differences between the treated and control groups post-treatment. In this sense, our selection of control industries is conservative.

²⁹ The input-output account data from BEA provides information at the detailed industry level for only 2007 and 2012. See <https://www.bea.gov/industry/input-output-accounts-data>.

³⁰ <https://www.federalregister.gov/documents/2008/07/15/E8-16011/current-good-manufacturing-practice-and-investigational-new-drugs-intended-for-use-in-clinical>.

soaps. Case 2 in [Figure 3](#) shows that the RegIndex of pharmaceutical and medicine manufacturing experience a slow upward trend similar to that of other chemical manufacturing before 2008. However, the RegIndex of pharmaceutical and medicine manufacturing increased dramatically after the FDA took regulatory actions in 2008, while RegIndex for other chemical manufacturing is essentially flat in the post treatment period. In contrast, RegData cannot distinguish the treated from the control groups, as can be seen from [Figure IA.2](#). The likely reason is that FDA guidance may not be immediately reflected in the CFR.

Case 4: Regulation of Hospitals. The fourth and final case study used for the validation of RegIndex focuses on the healthcare industry. The Patient Protection and Affordable Care Act (ACA) is a landmark U.S. federal statute enacted in 2010 under President Barack Obama. The ACA represents the U.S. healthcare system's most significant regulatory overhaul and expansion of coverage since the enactment of Medicare and Medicaid in 1965. At the time of its passing, hospitals faced significant pressure in coping with the new regulatory changes and dealing with insurers, all features that should correspond with higher levels of RegIndex.

For this case study, we regard hospitals (NAICS 622) as the treated group industry and use animal hospitals (NAICS 54194) as the control industry establishments. Case 4 in [Figure 3](#) validates the parallel trends assumption for RegIndex in hospitals and animal hospitals prior to 2010, which is once again suggestive of this exercise being informative. As can be seen from the figure, after 2010, the RegIndex of hospitals increases in both absolute terms and relative to the control group, appropriately tracing the heightened regulations imposed on the treated group relative to the controls. In this case, where the ACA is clearly mapped in the CFR, imposes heightened regulation, and separates treated and control industries very differently, supply-side measures are likely to also identify the regulation well. Indeed, we observe in the [Figure IA.2](#) that the RegData measure can also identify increased regulatory restrictions for hospitals from animal hospitals.

Finally, Table IA.4 reports the statistical significance of the graphical evidence of the cases in [Figure 3](#), and Figures IA.3 and IA.4 show that the results are robust to using the medium and broad versions of RegIndex.

2.5.3 State RegIndex and state voting for the Republican Party.

More indirect dimensions of the data may also be informative of the validity of the RegIndex methodology. Political parties support different approaches to regulation and intervention ([Peltzman 1998](#); [Mian, Sufi, and Trebbi 2010](#)). In the United States, Republican administrations make limiting the government burden on firms an explicit goal. Given the presence of substantial leeway at the state level in creating state-specific

regulatory environments (e.g., in the case of the insurance industry or state banking for instance, see Agarwal et al. (2014)), one would expect to see lower levels of RegIndex in Republican-controlled or Republican-leaning constituencies. This sanity check is illustrated below.

We begin by estimating state-specific RegIndex averages, conditional on the state's industry composition. That is, we extract state fixed effects in each year controlling for industry fixed effects, and recover the conditional mean RegIndex for each one of the 50 states and the District of Columbia. All establishments are weighted by their total wage payment and the sampling weights assigned by the OEWS survey. Industry is defined at the NAICS 6-digit level.

Figure 4 reports the heat map of state-specific RegIndex averages in 2014. States with the highest RegIndex include Democratic Party-leaning Vermont, Connecticut, Delaware, and Massachusetts, while states with the lowest RegIndex include Republican strongholds Alabama, Louisiana, North Dakota, and Mississippi. Comparing our state RegIndex with the state RegData, which counts for restrictive words in state regulatory texts since 2017,³¹ a notable difference is that state RegData is heavily related to the number of businesses in the state. For instance, states with the highest RegData are California, New York, New Jersey, Ohio, Illinois, and Texas, while states with the lowest RegData are South Dakota, Idaho, North Dakota, Alaska, and Montana (see Figure IA.5). States' number of establishments (from the Census SUSB) in 2017 explains 62% of the variation in the 2017 state RegData, where the coefficient has a t -statistics of 8.74. In contrast, states' number of establishments explains only 1% of our state RegIndex where the t -statistics of the coefficient is -0.74 .

More systematically, state RegIndex averages correlate negatively with a state's political inclination to vote for the Republican Party. As an illustration, we consider states' Republican vote shares in the 2016 presidential election (Donald Trump vs. Hillary Clinton), the 2016 House elections, and the 2018 Senate elections. Table 2 shows that state-specific RegIndex is significantly and negatively related to the state Republican vote share in all three elections.³²

2.5.4 Evidence on publicly traded firms' RegIndex. Finally, we validate our RegIndex measure using publicly traded firms. Motivated by

³¹ RegData starts to count restrictive words in state regulatory texts for 16 states in 2017, 9 additional states in 2018, 18 additional states in 2018, 3 additional states in 2020, and 3 additional states in 2022. Analyzing RegData that covers the same state in multiple years reveals that state RegData is extremely stable over time, as state fixed effects explain over 99.6% of the variation of the pooled state RegData sample. Hence, for each state, we use the earliest available state RegData to represent the state's RegData in 2017. We download state RegData at https://quantgov-bulk-downloads.s3.amazonaws.com/State-RegData-Definitive-Edition_regulations.zip.

³² In Table IA.5, we further control for state RegData and state number of establishments in the regression, and we observe very similar results to those in Table 2.

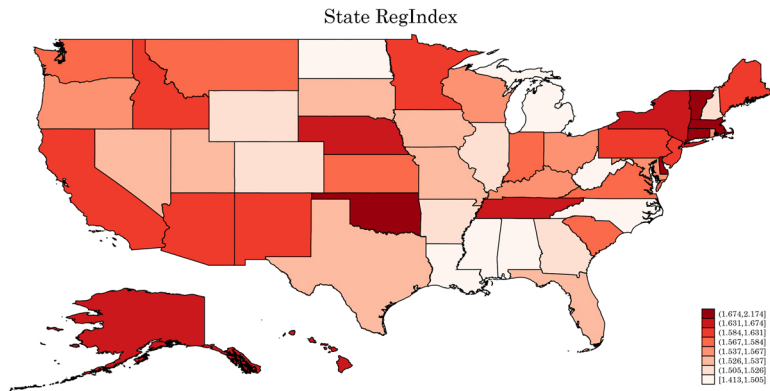


Figure 4
Regulation index across states

This figure plots the coefficients on state dummies in the following regressions based on about 1 million private establishments in the 2014 OEWS universe: $RegIndex_{i,t} = \alpha + \sum_{s \in States} \beta_s \times State_s + FE_{Ind} + \epsilon_{i,t}$. The coefficient α shows the RegIndex for the benchmark state, Alabama. The sum of coefficients $\alpha + \beta_s$ shows the RegIndex for the other 50 states (including the District of Columbia). All establishments are weighted by their total wage payment and their sampling weights assigned by the OEWS survey. Industry is at the NAICS 6-digit level.

Table 2
Validation: Vote share for Republican Party and RegIndex

	Republican Party Vote Share 2016 Presidential Election (1)	Republican Party Vote Share 2016 House Elections (2)	Republican Party Vote Share 2017–18 Senate Elections (3)
<i>State RegIndex</i>	−0.640*** (0.083)	−0.966*** (0.242)	−1.828*** (0.467)
<i>Constant</i>	1.502*** (0.132)	2.031*** (0.376)	3.408*** (0.741)
Observations	51	50	50
Adjusted R^2	0.411	0.315	0.119

This table reports the results of regressing state vote shares for the Republican Party in the 2016 presidential election, the 2016 House, and the 2017–18 Senate on the state 2014 RegIndex. See Figure 1 for the estimation of the state 2014 RegIndex. Standard errors are reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

the large literature examining the “disappearing IPO” puzzle (e.g., Gao, Ritter, and Zhu (2014)) and recent work by Ewens, Xiao, and Xu (2024) examining publicly traded firms bunching near the Sarbanes-Oxley Act’s (SOX) thresholds on public float, this subsection investigates whether RegIndex can capture changes in the corporate regulatory environment after initial public offerings.³³

To perform this exercise, we merge BLS establishments with publicly traded firms in the Compustat-CRSP database using a combination of EIN

³³ See Coates and Srinivasan (2014) for a review.

matching and fuzzy name matching.³⁴ Following BLS's suggested aggregation practices and prior literature, we represent each publicly traded firm's occupational employment using all of its matched establishments surveyed in the current year and the previous 2 years.³⁵ Importantly, using the establishment unique identifier, we are able to obtain publicly traded firms' occupational employment before they become public and enter the Compustat-CRSP merged database. Following Equation (3), we then measure each publicly-traded firm's RegIndex as the percentage of its labor spending on regulation-related tasks, and track the firm's RegIndex before and after the IPO.³⁶ Our final sample requires firms to be incorporated in the United States and have a RegIndex measure. The sample ends up covering 1,024 IPO events from 2002 to 2014.

Our specification regresses a firm's RegIndex on indicators of the IPO event year from -3 to 3 , with the year prior to the IPO as the benchmark, while controlling for year and firm fixed effects. Column (1) of Table 3 shows no significant pre-trend in RegIndex before the IPO. Instead, we detect a significant increase in a firm's RegIndex after the IPO, consistent with the hypothesis that being publicly traded exposes firms to a regulatory compliance burden higher than does being private.³⁷ The economic magnitudes are also meaningful. Two years after the IPO, a firm's RegIndex increases on average by about 8% ($= 0.151/1.798$) of the pre-IPO level at $t - 1$.

One concern about the exercise might be that the raw RegIndex may not account for changes in industry-level regulation for IPO firms. For instance, it is possible that regulation may tighten for all firms in a sector, private and public, after the industry undergoes a major wave of IPOs. We mitigate this concern by subtracting from the IPO firms' RegIndex the industry-year average of all publicly traded firms' RegIndex.³⁸ Column (2) shows that this demeaned RegIndex exhibits a pattern similar to the original RegIndex, suggesting that our IPO results do not appear to be entirely driven by industry trends. For completeness, Table IA.6 shows that the results are similar if one employs the medium and broad versions of RegIndex.

³⁴ See also Zhang (2019).

³⁵ Because the OEWS program surveys each establishment every 3 years, the BLS produces and publicizes aggregate statistics of occupational employment at the industry- or geographic-level using surveyed establishments from $t - 2$ to t ; see BLS technical notes at https://www.bls.gov/oes/oes_ocrch.htm.

³⁶ We obtain firm IPO dates from Jay Ritter's website.

³⁷ Notice that our goal for this subsection is simply to validate whether our RegIndex measure can produce findings in line with prior literature rather than fully establishing a causal effect (see Bernstein 2015 for a discussion of identification).

³⁸ We use the NAICS 4-digit code from Compustat to assign a publicly traded firm's industry.

Table 3
Validation: RegIndex before and after IPO

	RegIndex (1)	RegIndex (Industry Adjusted) (2)
IPO_{t-3}	0.047 (0.064)	0.059 (0.063)
IPO_{t-2}	-0.017 (0.037)	-0.003 (0.036)
IPO_t	0.045 (0.031)	0.039 (0.031)
IPO_{t+1}	0.135*** (0.047)	0.108** (0.046)
IPO_{t+2}	0.151*** (0.052)	0.111*** (0.051)
IPO_{t+3}	0.153*** (0.053)	0.099* (0.051)
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	5,164	5,164
Adjusted R^2	0.701	0.621

This table reports the results of regressing a publicly traded firm's RegIndex on event year indicators during the 3 years before and 3 years after it goes for the initial public offering (IPO) in the United States from 2002 to 2014. We obtain the firms' IPO year from [Jay Ritter's](#) website. We treat the year before IPO ($t - 1$) as the benchmark year, which is omitted in the regression. The mean RegIndex of IPO firms at $t - 1$ is 1.798. In column (1), we compute a publicly traded firm's RegIndex by averaging its establishments' RegIndex weighted by its establishments' total labor costs. In column (2), we demean the firm's RegIndex by subtracting the average RegIndex of its industry in the year, where industry for publicly traded firms is the NAICS 4-digit obtained from the Compustat database. Section 2.5.4 provides more details. All regressions control for firm and year fixed effects. All standard errors are clustered at the firm level and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Finally, we examine how RegIndex relates to two other prominent measures of publicly traded firms regulatory burden. Table IA.7 reports that our measure is moderately, but consistently correlated with the measure from [Armstrong, Glaeser, and Hoopes \(2025\)](#) (correlations of 0.240–0.281 depending on the versions of RegIndex) and the measure from [Kalmenovitz \(2023\)](#) (correlations of 0.165–0.182).

3. Regulatory compliance costs and business size

This section examines a crucial property of RegIndex_{it} and addresses the paper's main question. It will present evidence that regulatory compliance costs fall disproportionately on medium-sized businesses relative to others.

To explore economies of scale in regulatory compliance costs, we focus on the sign and magnitude of the derivative of the regulatory index with respect to firm (or establishment) employment, $\frac{\partial \text{RegIndex}_{it}}{\partial L_{it}}$.³⁹ We present both estimates for the whole U.S. economy and industry-specific estimates

³⁹ For simplicity of exposition and with a limited abuse of notation, we will refer to L_{it} as “employment” (as opposed to the proper total employment given by the sum $L_{it} + R_{it}$). Given the magnitudes that we report in this article, this approximation is warranted, and it makes both exposition and analysis much clearer. In the empirical analysis, for accuracy, we employ $L_{it} + R_{it}$.

that account for heterogeneity in regulatory regimes across different sectors.

As discussed earlier, economies of scale in regulatory compliance are a key feature of any regulatory architecture. Diseconomies of scale introduce a potential deterrent to firm growth, pushing firms to operate below their efficient scale of production. Regulation may also introduce incentives toward concentration and might act as a barrier to entry, favoring large incumbents.⁴⁰ While the issue of returns to scale in regulation has received much attention in the Political Economy and Industrial Organization literature in some specific industries, to the best of our knowledge, this paper is the first to provide a comprehensive set of facts representative of the entirety of the U.S. economy.

The simple framework in Equation (1) allows one to explicit several factors driving economies (or diseconomies) of scale under a limited set of assumptions. There are at least two factors. First, in the model, fines imposed by regulators are a function of requirements imposed by the rules, $\bar{R}_{it}$. These standards are, in turn, a function of size, $\bar{R}_{it} = \bar{R}(L_{it})$. Importantly, it is plausible to hypothesize $\frac{\partial \bar{R}_{it}}{\partial L_{it}} \geq 0$.⁴¹ An example of a positive derivative is capital requirements imposed on large bank holding companies kicking in at several thresholds for total assets and as a function of the systemic importance of the financial institution (both measures correlate with employment). Another is the regulatory tiers discussed in Brock and Evans (1985). An example of a negative derivative is instead presented in Hopkins (1995), which shows that the smallest firms in his sample have paperwork and tax compliance costs (measured against turnover) about twice as large as those of the largest firms.

As a second factor, scale matters through the probability of inspection p_{it} . The inspection probability is naturally driven by government enforcement effort, which can be a function of size, $E_{it} = E(L_{it})$. One may plausibly posit $p_{it} = p(E_{it})$, with $\frac{\partial p_{it}}{\partial E_{it}} \geq 0$, and $\frac{\partial E_{it}}{\partial L_{it}} \geq 0$, where this derivative may be positive if larger firms have more weight in inspection protocols or negative if, for instance, smaller plants are easier/faster to inspect.⁴²

These considerations suffice to illustrate a theoretical ambiguity in the relationship between regulatory costs and scale. Using Equation (1) and

⁴⁰ Classic references are Stigler (1971); Peltzman (1976).

⁴¹ In addition, note that fines may be a function of the firm's covariates and size (in addition to the level of compliance exerted by the firm relative to a given standard required by the rules, $\bar{R}_{it}$). That is, one could posit a general $f_{it} = f(L_{it}, R_{it}, \bar{R}_{it})$ with $\frac{\partial f_{it}}{\partial L_{it}} \geq 0$, where a positive derivative case can arise if fines are designed to be more than proportional to firm size "to set an example," and negative if fines are capped by statutory limits/by the threat of litigation from large firms.

⁴² See Helland (1998) for a discussion related to the Environmental Protection Agency in the United States. See also Shimshack (2014) for a review of the evidence.

the discussion above, we have $R_{it} = \left(\frac{p(E_{it})\bar{R}_{it}}{w_{it}^r} \right)^{\frac{1}{2}}$, where $\frac{\partial R_{it}}{\partial L_{it}} \geq 0$ depending on the dominating force. Importantly, using the definition (2), it follows that the sign of $\frac{\partial \text{RegIndex}_{it}}{\partial L_{it}}$ is also ambiguous.

This ambiguity is borne out by the data. Figure 5 presents firm- and establishment-level nonparametric evidence of a non-monotonic relationship between total employment size and RegIndex. The dots in the graph represent averages for employment bins of [1, 2] employees, [3, 4], [5, 9], [10, 19], [20, 49], [50, 99], [100, 249], [250, 499], [500, 749], [750, 999], [1000, 1499], [1500, 2499], [2500, 3999], and above 4000. As firm or establishment employment increases regulatory costs per employee increase steadily until a firm size of around 500 workers, then regulatory compliance costs per employee start falling rapidly, indicating economies of scale. We estimate the percentage of labor costs for regulatory compliance for mid-sized businesses to be about 40–50% higher than that for the smallest businesses and about 10–20% higher than that for the largest ones.

The evidence from a parametric representation of the non-monotonicity is reported in Table 4, which includes the max and argmax of an inverted-U relationship between RegIndex and size for both firms and establishments. The coefficients of the parametric regression are significant and precisely estimated across all different specifications. The specifications in the table include different sets of fixed effects to assess the robustness of the finding: Year FE, Year×Industry FE, Year×Industry×State FE, and Year FE + Firm FE. The table reports a range for argmax in panel A of about 499–511 total employees for firms at the peak of regulatory compliance costs and an argmax in panel B of about 309–344 for establishments. As the average firm in the United States includes only 1.26 establishments from 2002 to 2014 according to the Census SUSB, it is not completely surprising that argmax aligns between panels A and B. Finally, Figure IA.6 shows that our finding holds for all three versions of RegIndex.

Figure 6 reports the same information by large sector aggregates, highlighting a degree of heterogeneity in the presence of regulatory scale economies. The pictures present evidence both at the firm level in panel A and the establishment level in panel B. Both figures show a non-monotonic relation between RegIndex and size within industries such as finance, retail, and other services. The non-monotonicity is less pronounced in others, such as manufacturing and utilities, as one would expect from the vast differences in sector specific regulations.

Figure IA.7 provides a visual representation of the RegIndex across employment bins each year from 2002 to 2014. While RegIndex increases for all size bins over time, the inverted-U shape appears highly stable over time, with some changes in the left tail of the size distribution (firms with

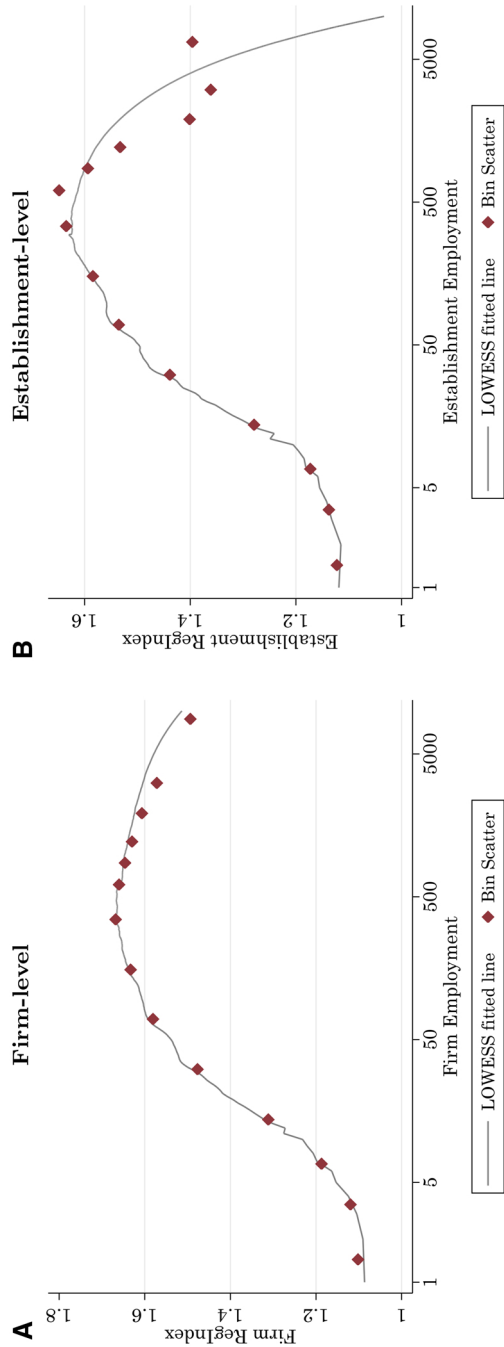


Figure 5

RegIndex and size

This figure plots the relation of RegIndex and employment for firms in panel A and establishments in panel B. The dots represent the average RegIndex in each employment bin, where the bins are [1, 2], [3, 4], [5, 9], [10, 19], [20, 49], [50, 99], [100, 249], [250, 499], [500, 999], [1000, 1499], [1500, 2499], [2500, 3999], and above 4000. The line represents the LOWESS smoothed fitted curve using the bandwidth of 0.05. The x-axis is on a log scale.

Table 4
Economies of scale for RegIndex
A: Firm level

	(1)	(2)	(3)	(4)	(5)
<i>Emp</i>	2.897*** (0.065)	2.920*** (0.065)	2.008*** (0.076)	1.935*** (0.068)	0.544*** (0.074)
<i>Emp</i> ²	-2.902*** (0.068)	-2.927*** (0.068)	-1.963*** (0.073)	-1.909*** (0.064)	-0.542*** (0.069)
max	1.961*** (0.028)	1.965*** (0.015)	1.782*** (0.018)	1.755*** (0.016)	1.517*** (0.018)
argmax	0.499*** (0.003)	0.499*** (0.003)	0.511*** (0.003)	0.507*** (0.003)	0.501*** (0.018)
Year FE	—	Yes	—	—	Yes
Year-Ind FE	—	—	Yes	—	—
Year-Ind-State FE	—	—	—	Yes	—
Firm FE	—	—	—	—	Yes
Observations	3,027,680	3,027,680	3,027,241	2,918,296	2,162,080
Adjusted <i>R</i> ²	0.007	0.007	0.378	0.412	0.597

B: Establishment level

	(1)	(2)	(3)	(4)	(5)
<i>Emp</i>	3.607*** (0.265)	3.788*** (0.217)	2.985*** (0.148)	2.902*** (0.134)	0.612*** (0.125)
<i>Emp</i> ²	-5.255*** (0.486)	-5.510*** (0.430)	-4.452*** (0.284)	-4.397*** (0.259)	-0.992*** (0.197)
max	1.823*** (0.041)	1.850*** (0.019)	1.724*** (0.016)	1.699*** (0.014)	1.444*** (0.016)
argmax	0.343*** (0.007)	0.344*** (0.007)	0.335*** (0.006)	0.330*** (0.005)	0.309*** (0.012)
Year FE	—	Yes	—	—	Yes
Year-Ind FE	—	—	Yes	—	—
Year-Ind-State FE	—	—	—	Yes	—
Establishment FE	—	—	—	—	Yes
Observations	3,362,824	3,362,824	3,362,418	3,255,415	2,194,239
Adjusted <i>R</i> ²	0.005	0.006	0.371	0.408	0.534

Panel A reports the results of regressing firms' RegIndex on their employment and employment squared, where RegIndex is shown as a percentage and employment in 1,000s. Panel B reports the results at the establishment level. All standard errors are double clustered by year and firm in panel A and at year and establishment in panel B. Industry is defined at the NAICS 6-digit level. Given a quadratic formula $y = ax^2 + bx + c$, the max is computed as $c - \frac{b^2}{4a}$, while the argmax is computed as $-\frac{b}{2a}$. The sample period is from 2002 to 2014. * $p < .1$; ** $p < .05$; *** $p < .01$.

less than 10 employees), which appears to become flatter from 2002 to 2014. A possible reason for this pattern is the relative reduction of the regulatory burden on small firms over this time period.⁴³

⁴³ An example arises from the Small Business Administration's (SBA) advocacy in favor of small firms. In the August 2002, Executive Order 13272 by President George W. Bush further implemented the Regulatory Flexibility Act of 1980, which is the act codifying the SBA. Over the period 2002–2009 one can find several instances in which the SBA Office of Advocacy actively lobbied to reduce compliance costs on small businesses. Examples include, in 2008, when the EPA finalized the Lead Renovation, Repair, and Painting Rule requiring contractors working in pre-1978 homes to be certified and follow specific lead-safe work practices. The rule was simplified in its requirement for small residential construction and renovation companies, lessening costs and complexity of compliance for small-business contractors. With the passage of the Small Business Jobs Act of 2010, the Office of Advocacy further strengthened. This, combined with the vast regulatory intervention after the 2009 financial crisis (e.g., the Dodd-Frank Act of 2010 and the ACA of 2010), likely contributed to a relative reduction in compliance costs for small businesses over 2010–2014.

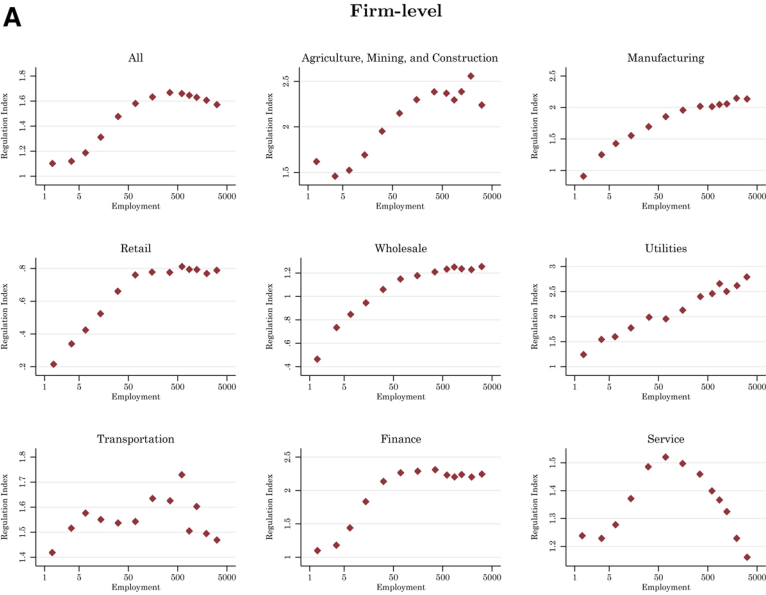


Figure 6
RegIndex and size by sector

Panel A shows the relation of RegIndex and firm employment in each NAICS 1-digit sector. Panel B shows the relation at the establishment level. Each dot represents the average RegIndex in each employment bin, where the bins are [1, 2], [3, 4], [5, 9], [10, 19], [20, 49], [50, 99], [100, 249], [250, 499], [500, 749], [750, 999], [1000, 1499], [1500, 2499], [2500, 3999], and above 4000. The x-axis is on a log scale.

3.1 Robustness

3.1.1 Within-occupation variation. One might be concerned that workers in the same occupation may in practice perform more regulation-related tasks in small firms than in mid-sized firms. For instance, small firms may hire non-regulation-related occupations to cover non-regulation-related tasks and also cover some regulatory compliance tasks. To assess how firms' requirements on regulation-related tasks vary with their size for a given occupation, we investigate firms' skill requirements in their job posting descriptions using over 14 million job postings during 2010–2014 from the Burning Glass Technologies (BGT) data.

The BGT data provide 17,420 skills extracted from the millions of job posting descriptions. For each skill, the data also provide a skill definition, usually one sentence, like the task statements in the O*NET data. We identify a BGT skill as “regulation-related” using the same procedure that identified regulation-related tasks in [Section 2.3](#). This procedure yields

For instance, in the case of the Affordable Care Act of 2010, the threshold of 50 employees for the application of the Employer Mandate is an area where the Office of Advocacy actively intervened, including by helping simplify reporting requirements, facilitating tax compliance for firms around the threshold, and advocating for tax credits for health care for affected business just above the regulatory “cliff.”

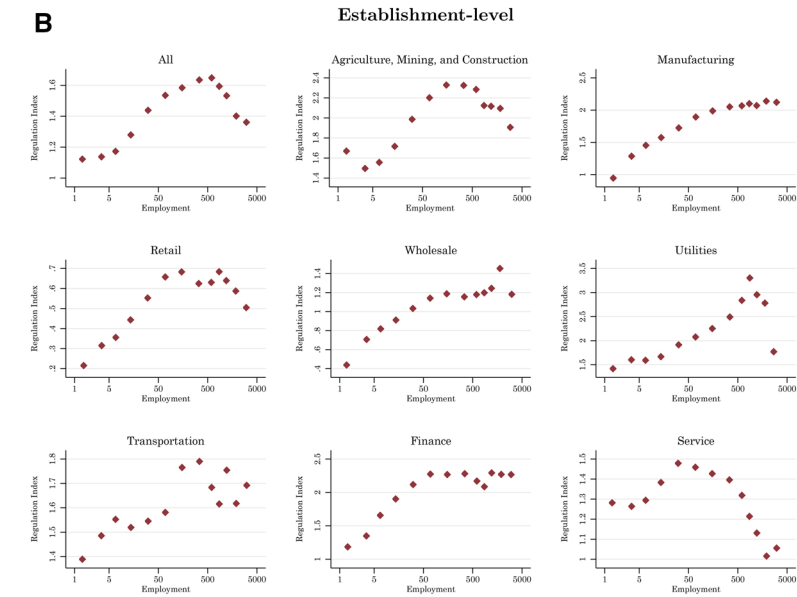
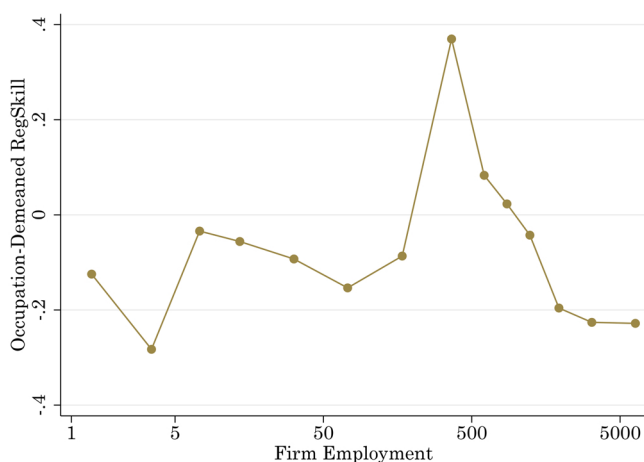


Figure 6
(Continued)

523 regulation-related skills in the BGT data. Averaging the regulation-related dummies for all skills within a job posting generates a continuous regulation-relatedness measure for each job posting. Finally, we name match BGT firms to OEWS firms and regress the regulation-related measure of job postings on the posting firms' employment and employment squared. Because the regulation-related measure is at the job posting level and varies within an occupation, we can now explore small and large firms' requirement on regulatory compliance when hiring the same occupation in the same year. This allows, for example, one to investigate whether the regulatory skill requirements for an electrical engineer are particularly high for an engineer working in a small firm relative to a large one.

Figure 7 plots the average of the demeaned regulation-relatedness measure for job postings of occupations with positive RTI in each firm employment bin, where each job posting's regulation-relatedness measure is demeaned by subtracting the average regulation-relatedness measure of all job postings in that occupation in that year. The figure shows a similar inverted-U shape, where firms with around 500 employees have greater requirements for performing regulatory compliance tasks than small and large firms when hiring the same occupation in the same year.

Table 5 shows the results more formally by regressing each job posting's regulation-relatedness measure on firm size while controlling for Year×Occupation fixed effects. Column (1) provides evidence that

**Figure 7****Within-occupation regulatory compliance requirement and firm size**

This figure plots the demeaned regulatory compliance skill requirement measure for job postings in each firm employment bin, where a job posting's regulatory compliance skill requirement measure is demeaned by subtracting its mean within an SOC 6-digit occupation and year. See [Section 3.1.1](#) for details. The bins are [1, 2], [3, 4], [5, 9], [10, 19], [20, 49], [50, 99], [100, 249], [250, 499], [500, 749], [750, 999], [1000, 1499], [1500, 2499], [2500, 3999], and above 4000. The x-axis is on a log scale.

firms' demand for regulatory compliance tasks also exhibits an inverted-U relationship with firm size, though with the peak at around 800 employees. Specifically, for the same occupation, mid-sized firms appear to require more regulatory compliance skills than small and large firms. Columns (2) and (3) further show that mid-sized firms require more regulatory compliance skills than small and large firms regardless of hiring regulation-related occupations ($RTI > 0$) or non-regulation-related occupations ($RTI = 0$).

Overall, these results suggest that our finding on the inverted-U relationship between RegIndex and firm size is a conservative estimate. The relationship between firms' regulatory compliance costs and their size would show an even more pronounced inverted-U shape if we were to fully account for within-occupation variation in regulatory compliance requirements.

3.1.2 Capital. Next we consider whether the findings so far may be sensitive to (or even entirely driven by) small, medium, and large firms' heterogeneous use of labor and physical capital in complying with regulation.

As discussed in [Section 2](#), precisely estimating firms' use of capital for regulatory compliance purposes is challenging, mostly due to the lack of detailed data on various capital user costs at the micro level in the OEWS. Nonetheless, following [Caunedo, Jaume, and Keller \(2023\)](#) and focusing

Table 5
Regulatory compliance within occupation

Sample:	Dependent Variable: Regulatory Compliance Skill Requirement		
	All Occ (1)	Occ (RTI > 0) (2)	Occ (RTI = 0) (3)
<i>Emp</i>	0.722*** (0.204)	0.536*** (0.168)	0.858*** (0.290)
<i>Emp</i> ²	-0.445*** (0.131)	-0.341*** (0.102)	-0.520*** (0.181)
max	1.558*** (0.092)	2.073*** (0.071)	1.188*** (0.128)
argmax	0.811*** (0.048)	0.787*** (0.048)	0.825*** (0.059)
Year-Occ FE	Yes	Yes	Yes
Observations	14,052,988	5,878,039	8,174,949
Adjusted R ²	0.139	0.138	0.119

This table reports the relationship of firms' requirements for regulatory compliance skills in job postings and their size within an occupation. The dependent variable is the job posting's regulatory compliance skill requirement constructed as following: We first name match firms in our OEWS sample to firms in the Burning Glass Technologies (BGT) data, which provide 14 million job postings from 2010 to 2014 (the overlapping period between Burning Glass and our OEWS sample) where each job posting describes a set of required skills for the job. We next label each skill as performing regulation-related tasks by applying our methodology in Section 3 to the textual contents for the skill in the BGT data. A job posting's regulatory compliance skill requirement is the share of all skill requirements in the job posting that is about performing regulation-related tasks (in percentage). We regress the job posting's regulatory compliance skill requirement on the posting firm's employment (in thousands) and the squared of the employment, while controlling for year-occupation fixed effects. Column (1) reports results for the full sample with all SOC 6-digit occupations. Column (2) requires results for the subsample of occupations that have positive regulation-task intensity (RTI) in our definition in Section 2.3, while column (3) reports the results for the subsample of occupations with RTI equal to 0. All standard errors are clustered at the firm (EIN) level and reported in parentheses. Given a quadratic formula $y = ax^2 + bx + c$, the max is computed as $c - \frac{b^2}{4a}$, while the argmax is computed as $-\frac{b}{2a}$. See Section 3.1.1 for details. * $p < .1$; ** $p < .05$; *** $p < .01$.

on O*NET tools and capital equipment requirements of occupations, we are able to construct an approximate measure of share of regulatory compliance costs out of the total labor and capital costs in Section 2. This measure, $\text{RegIndex}_{i,t}^{\text{tot}}$, is employed in replicating Figure 5.

Figure 8 shows how labor-based $\text{RegIndex}_{i,t}$ and total-input-based $\text{RegIndex}_{i,t}^{\text{tot}}$ compare in terms of their relationship with firm and establishment size. We again observe a distinctive inverted-U shape for $\text{RegIndex}_{i,t}^{\text{tot}}$, proving the robustness of our main finding. There are, however, some important differences relative to $\text{RegIndex}_{i,t}$. The inverted-U shape traced by using $\text{RegIndex}_{i,t}^{\text{tot}}$ coincides with $\text{RegIndex}_{i,t}$ for small firms and establishments, as expected given the limited role of physical capital for very small units. However, the $\text{RegIndex}_{i,t}^{\text{tot}}$ curve remains consistently lower than $\text{RegIndex}_{i,t}$, indicating, as mentioned above, that labor inputs play a disproportionately important role relative to capital inputs in regulatory compliance. Further, the distance between the two curves widens as firm and establishment increase in size, indicating that capital

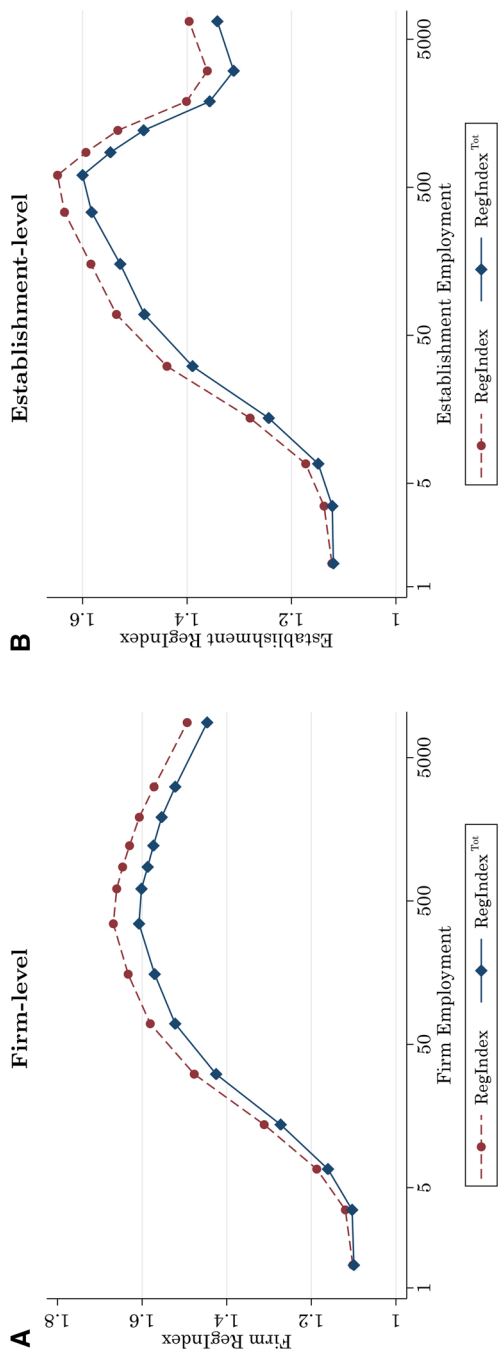


Figure 8
Total-input-based RegIndex and size

This figure plots the relation of labor-based RegIndex (baseline), total-input-based RegIndex (including capital) and employment for firms in panel A and establishments in panel B. See [Section 2.3](#) for definitions of RegIndex and RegIndex^{Tot}. The dots represent the average RegIndex in each employment bin, where the bins are [1, 2], [3, 4], [5, 9], [10, 19], [20, 49], [50, 99], [100, 249], [250, 499], [500, 749], [750, 999], [1000, 1499], [1500, 2499], [2500, 3999], and above 4000. The x-axis is on a log scale.

plays an increasingly relevant role in regulatory compliance cost savings as firm grow in size.⁴⁴

3.1.3 Outsourcing. Firms also hire outside compliance professionals to comply with regulation. In this subsection, we explore whether outsourcing may be a major source of bias for our main findings.

How outsourcing might affect our main results is a priori unclear. On one hand, small firms may have greater incentives to outsource compliance tasks due to limited resources and lack of economies of scale in in-house compliance. On the other hand, large firms may prefer outsourcing due to an increased focus on streamlining operations and allocating in-house workforce to “core competencies” (e.g., [Goldschmidt and Schmieder 2017](#)). Moreover, if outsourcing itself involves fixed costs—such as those associated with searching for service providers and setting up business with external suppliers—larger firms may find it even more economical than small firms to outsource compliance.

While we do not observe firm-to-firm contracts for regulatory compliance outsourcing in our data, this subsection provides two sets of facts suggesting that outsourcing is unlikely to qualitatively affect our main finding. We first leverage two major industry survey reports in order to assess how small and large firms outsource in practice. A first report is from the [National Association of Manufacturers \(2014\)](#) and provides the likelihood for small, medium, and large firms to hire outside advisers in complying with federal regulations for the manufacturing sector. Chart 3 of [National Association of Manufacturers \(2014\)](#) shows that this likelihood is smallest for small firms; 58%, increasing to 70% for medium firms; and 82% for large firms. Based on this monotonic relation, heterogeneous outsourcing appears unlikely to overturn our finding that RegIndex is lower for small firms than for medium and large firms in [Figure 6](#).⁴⁵ This finding is also consistent with [Chayes and Chayes \(1985\)](#) showing that in-house and outsourced regulatory spending complement each other.

A second report from the [Securities Industry Association \(2006\)](#) shows that, while outsourcing spending accounts for only 2.8% of financial sector firms’ total compliance-related spending (as compared to 93.9% of in-house labor spending), outsourcing spending accounts for a larger fraction relative to in-house labor spending in small firms than in large firms. Calculations based on [Figure 3b](#) of [Securities Industry Association \(2006\)](#) show that outsourcing spending is about 16.4%, 3.2%, and 2.2% of the in-house compliance-related labor spending for small, medium, and large

⁴⁴ It can also be possible that the regulation-related capital expenditure is more difficult to consolidate than labor costs, reducing the economies of scale for larger firms.

⁴⁵ One limitation is that this report only shows the extensive margin, but not the intensive margin, for small, medium, and large firms’ compliance outsourcing.

firms, respectively. One can further evaluate the effect of outsourcing spending by computing an outsourcing-adjusted RegIndex by adding outsourcing costs to both the numerator and denominator of Equation (4). Figure IA.8 shows that such an adjustment does not overturn our main findings either.

A second set of facts can be inferred from a more comprehensive assessment of how accounting for costs paid to external compliance specialists affects our main findings. We construct a measure of outsourcing costs for establishments and firms based on the following intuition. Suppose we consider a certain geographic market, say a U.S. county, and focus on the employment distribution in the county across small, medium, and large establishments. Suppose further that the market for compliance service providers is localized in nature. Under these assumptions, we can test whether small establishments utilize more outsourcing services than medium or large establishments. That is, if one were able to compare two counties similar in everything except that one has a larger share of employment in small establishments than the other, we should observe a higher presence of outsourcing service providers in the former relative to the latter.

With this idea in mind, we develop a co-location estimator across U.S. counties to assign outsourced compliance costs to establishments in different establishment size bin of the inverted-U shape plot. Our assumption behind this estimation is that, in the period from 2002 to 2014, outsourcing is largely provided via on-site services and that establishments of different sizes have a different utilization rate of such services.⁴⁶

Operationally, our estimates focus on five major compliance service providers: legal, accounting and tax, payroll, IT, and management consulting services. For all these outside service providers, we observe detailed wage bill information. As these providers' production costs are ultimately paid by client establishments (i.e., businesses that outsource their compliance), we can conservatively use these providers' labor spending on regulation-related tasks to proxy for the total outsourced regulatory compliance client spending each year.⁴⁷ We define the outsourcing services intensity in each county, $OutSource_{c,t}$, as the total labor spending on regulation-related tasks from outsourcing-providing establishments in county c divided by all other establishments' total labor spending

⁴⁶ As an example, Charoenwong et al. (2024) show that technology-driven compliance (that is, RegTech) in finance arises after SEC amendments in 2014.

⁴⁷ The five compliance outsourcing industries are 5411 (Legal Services), 5412 (Accounting, Tax Preparation, Bookkeeping, and Payroll Services), 5415 (Computer Systems Design and Related Services), 5416 (Management, Scientific, and Technical Consulting Services), and 5419 (Other Professional, Scientific, and Technical Services).

in c .⁴⁸ We then compute each county's establishments' size distribution, $Share_{j,c,t}$, as the share of labor spending from each of the 14 size bins described in Section 3, where j represents each size bin. Finally, we regress $OutSource_{c,t}$ on the 14 $Share_{j,c,t}$ without imposing a constant term, while controlling for major county characteristics including the county's total employment, state and urban fixed effects, and year fixed effects.⁴⁹ This regression generates 14 coefficients for the size bins representing the relative usage of compliance outsourcing for different establishment sizes. A higher coefficient here proxies for a higher intensity of using compliance outsourcing for establishments in that bin relative to those in other bins.⁵⁰ Figure IA.10 plots these estimates. To quantify the impact of outsourcing, we estimate the outsourced compliance costs in addition to in-house compliance costs, $RegIndex^O$, for establishments in each size bin.

We present two versions for robustness. Our first version assumes that establishments only use outsourcing service providers within their county, as per the co-location estimation's assumption. Our calculation procedure is the following. First, denote $RegIndex_{j,c,t}^O$ as the average outsourced compliance costs normalized by total labor costs for establishments in size bin j and county c . Denote β_j as the relative outsourcing intensity for each size bin j estimated above; we can express $RegIndex_{j,c,t}^O = x_{c,t} \times \beta_j$, where $x_{c,t}$ is a benchmark value that we will solve for later. Second, note that county c 's observed outsourcing intensity, $OutSource_{c,t}$, is the average of $RegIndex_{j,c,t}^O$ weighted by the total labor costs of establishments in size bin j and county c at t , $w_{j,c,t}$. Third, solving for $x_{c,t}$ based on the above setting results in $RegIndex_{j,c,t}^O = (OutSource_{c,t} \times \frac{\sum_s \beta_s \times w_{s,c,t}}{\sum_s w_{s,c,t}}) \times \beta_j$,⁵¹ which we assign as the outsourced compliance costs for each establishment i in size bin j and county c at t . Finally, adding establishment i 's estimated $RegIndex_{i,t}^O$ to its baseline (in-house) $RegIndex$

⁴⁸ As a first pass check, we plot in Figure IA.9 the relation between $RegIndex$ and establishment size in three subsamples of counties with low, medium, and high outsourcing abundance, defined by terciles of $OutSource_{c,t}$ in the year. We observe that small establishments in counties with low outsourcing abundance counties still have lower $RegIndex$ than mid-sized establishments in those counties, demonstrating the robustness of our inverted-U shape results to the effect of outsourcing.

⁴⁹ We obtain each county's 2010 urban population share from the Census website at <https://www.census.gov/programs-surveys/geography/guidance/geo-areas/urban-rural/2010-urban-rural.html>. A county is classified as urban if more than half of its population resides in urban areas, as opposed to rural areas.

⁵⁰ Establishments from the largest size bin are sparsely distributed across counties, making the outsourcing intensity estimate for that bin volatile. We keep the 14 bins in this estimation to be consistent with our baseline specification. Nevertheless, our focal test result on how outsourcing affects small versus mid-sized establishment $RegIndex$ is very similar with and without including the largest size bin in the estimation. Given the volatile results for the largest size bin, we exclude this bin in the relevant plots.

⁵¹ In particular, we solve for $x_{c,t}$ from $OutSource_{c,t} = \sum_j \frac{RegIndex_{j,c,t}^O \times w_{j,c,t}}{\sum_j w_{j,c,t}} = \sum_j \frac{x_{c,t} \times \beta_j \times w_{j,c,t}}{\sum_j w_{j,c,t}}$.

results in the establishment's RegIndex with outsourced compliance costs, $RegIndex_{i,t}^{I+O}$.

Our second version assumes that all establishments use services from compliance outsourcing firms nationwide. In this case, establishments in each size bin, regardless of location, have the same $RegIndex^O$ in a year, which we back out through the same procedure as described above but use the nationwide outsourcing intensity (instead of the county-specific $OutSource_{c,t}$) in each year.

These two versions are based on polar assumptions about the local constraints of regulatory compliance outsourcing and thus provide bounds on the impact of outsourcing costs. We also compute the two versions at firm level (EIN) by averaging an EIN's establishments' $RegIndex_{i,t}^{I+O}$ weighted by the establishments total labor costs.

Figure 9 plots the two versions of firm-level $RegIndex_{i,t}^{I+O}$ and the baseline RegIndex by size bins in panel A and similarly at the establishment level in panel B. We observe that, while outsourcing costs disproportionately increase RegIndex for small firms, it does not change the qualitative inference that small firms pay proportionately lower regulatory compliance costs than mid-sized firms. This finding mitigates the concern that compliance outsourcing may overturn the left-hand side of our inverted-U shape results.

One final potential concern is that only small outsourcing service providers may focus on serving local clients, while large outsourcing providers may serve clients nationwide, making large outsourcing providers less suitable for our co-location estimation. If small outsourcing providers serve smaller firms more, we are likely to underestimate the outsourcing intensity of small firms. To mitigate this concern, we rerun our co-location estimation using only small outsourcing providers in the county, while assuming that large outsourcing providers affect all establishments' RegIndex equally. To the extent that large outsourcing providers serve larger establishments more, this re-estimation is likely to overestimate small establishments' outsourcing intensity relative to larger establishments, providing an upper bound for outsourcing's effect on overturning our main results. We define small outsourcing providers as those with 20 or fewer employees. Panels C and D of Figure 9 show that accounting for costs paid to small outsourcing providers, $RegIndex_{i,t}^{I+O}$ is still lower for small firms than mid-sized firms. We also experiment with different cutoffs for defining small outsourcing providers such as using 10 or 50 employees, and we find very similar results (see Figure IA.11). Finally, our main findings are robust if we rerun the estimation using medium or broad versions of RegIndex throughout (see Figure IA.12).

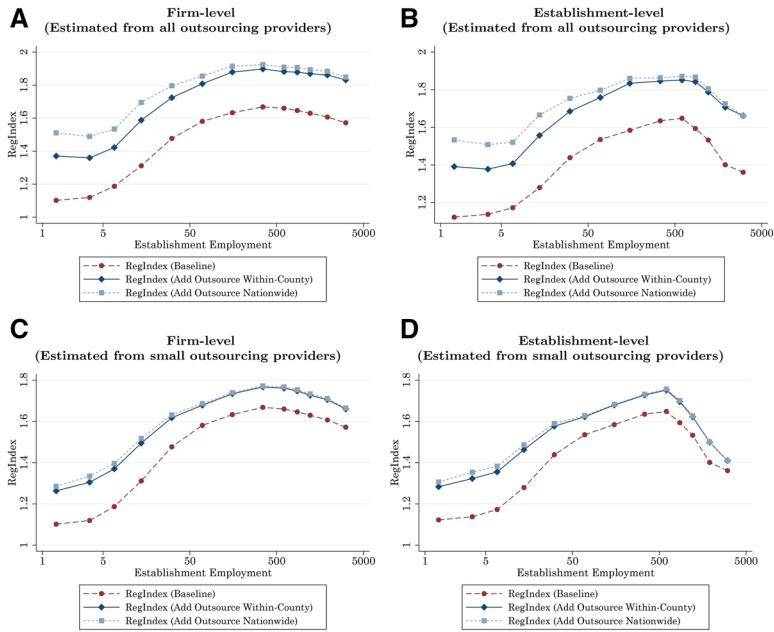


Figure 9
Outsource-adjusted RegIndex and size

This figure plots the relation of in-house-based RegIndex (baseline), two versions of outsource-adjusted RegIndex and employment for firms in panels A and C and establishments in panels B and D. In panels A and B, we include compliance costs paid to all compliance outsourcing firms, whereas in panels C and D, we include compliance costs paid to small outsourcing firms with 20 or fewer employees. Each panel reports two versions of outsource-adjusted RegIndex, constructed assuming that firms use compliance outsourcing services from local providers within the county and from nationwide providers, respectively. [Section 3.1.3](#) provides the detailed definitions and estimation procedures. See [Section 2.3](#) for the definition of the baseline RegIndex. The dots represent the average RegIndex in each employment bin, where the bins are [1, 2], [3, 4], [5, 9], [10, 19], [20, 49], [50, 99], [100, 249], [250, 499], [500, 749], [750, 999], [1000, 1499], [1500, 2499], [2500, 3999], and above 4000. The x-axis is on a log scale.

Overall, the evidence in this subsection suggests that accounting for outsourcing appears unlikely to overturn the key implications of our analysis.

3.1.4 Multi-state regulations. As an final robustness check, we examine whether firms' heterogeneous exposure to multiple state regulations plays an outsized role in driving the inverted-U shape. In particular, one might be concerned that some of the non-monotonicity in [Figure 5](#) may originate from a specific mixture of single-state firms, exposed to limited regulation, and medium-large multi-state firms, facing the extra complexities of complying with multiple states' regulations. Looking at single-state versus multi-state firms in isolation may help assuage this concern, especially if similar non-monotonicities are present for all types of firms.

We classify firms as follows. First, we identify single-unit establishments (based on a flag measure provided by the BLS) as the obvious group for single-state firms. These establishments account for over 80% of all establishments in our final sample. Second, if the establishment's parent EIN spans multiple states, we classify the firm (and establishments) as obviously part of a multi-state business. Third, we consider the residual group of firms (not belonging to the first or the second group) as likely to be single-state.⁵²

Figure 10 plots the relationship between RegIndex and firm size for single-unit firms, likely-single-state firms, and multi-state firms. We observe that both single-unit firms and likely-single-state firms show a strong inverted-U shape—clearly not a flat relationship. This is reassuring, as, for example, it rules out that our inverted-U shape result may be solely the result of omitted heterogeneity in the regulatory compliance profiles of multi-state versus single-state firms.

The RegIndex for multi-state firms shows two interesting features. First, multi-state firms show a higher level of RegIndex compared to single-state firms of the same size, confirming that the complexity of multi-state regulation increases firms' regulatory compliance costs. Second, we observe the inverted-U shape for firm-level RegIndex for multi-state firms, but a largely increasing shape for establishment-level RegIndex for multi-state firms. These differences are consistent with multi-state firms concentrating their regulatory compliance costs within few establishments rather than decentralizing their regulatory compliance, possibly with the goal of achieving economies of scale.

4. Mechanisms and Implications

This section presents empirical evidence on the mechanisms behind the key results of Section 3. We begin by investigating economies of scale, probably the most intuitive driver of the decreasing component of the inverted-U shape traced by Figure 5. Section 4.1 shows evidence consistent with the economies of scale mechanism for large firms to have a lower RegIndex than mid-sized firms. To understand why small firms have a lower RegIndex than mid-sized firms, we engage in a more complex exercise in Section 4.2, designed to decouple the different roles of regulatory requirements and of enforcement in driving compliance costs across different employment sizes, finding evidence consistent with regulatory tiering as a potential additional mechanism behind our findings.

⁵² Some of them may be multi-state if a firm presents multiple EINs across states or if the OEWS sampling misses certain EINs in other states, but most appear to be single-state firms with multiple establishments within a state.

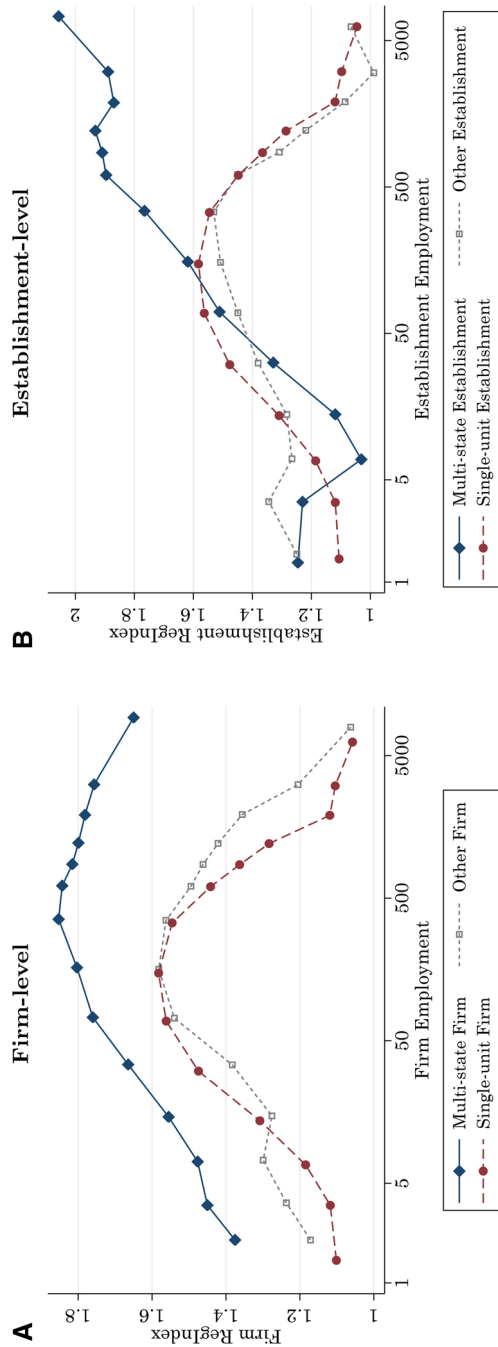


Figure 10
RegIndex and size: Multi-state vs. single state firms

This figure plots the relation of RegIndex and employment for firms in panel A and establishments in panel B. In each panel, we plot RegIndex for multi-state firms, single-unit firms, and other firms. See more details in Section 3. The dots represent the average RegIndex in each employment bin, where the bins are [1, 2], [3, 4], [5, 9], [10, 19], [20, 49], [50, 99], [100, 249], [250, 499], [500, 999], [1000, 1499], [1500, 2499], [2500, 3999], and above 4000. The x-axis is on a log scale.

4.1 Fixed costs

We begin by exploring what is intuitively a simple, but important economic mechanism behind the nonlinear relation between compliance costs and firm size: fixed costs in regulatory compliance.

Evidence that part of the increasing returns to scale in regulatory compliance (i.e., the downward part of the parabola traced in [Figure 5](#)) may derive from fixed costs can be inferred from the growing presence of regulatory compliance specialists as firm size increases. Compliance specialists may in fact offer more efficient handling of regulatory requirements, but may be affordable only for firms reaching a certain scale. The OEWS data allows us to carefully perform this check through its detailed occupation information.

For each firm's RegIndex, we compute the fraction of RegIndex that is performed by regulatory specialists, where specialists are defined using the top 25 occupations with the highest regulation-task intensity (see RTI in [Table IA.2](#)). The higher the share of RegIndex performed by specialists, the more the firm concentrates its compliance in a dedicated subset of its workforce.

The relationship between regulatory compliance specialization and size is reported in [Figure 11](#), which captures the increasing firms' reliance on regulatory specialists as their size increases. The figure shows that specialization of compliance is almost monotonically increasing with employment. This indeed suggests that larger firms pay the fixed costs of hiring compliance specialists, who can then consolidate regulatory compliance tasks and perform them more efficiently. The increment is quantitatively sizable, as the spending in regulatory compliance through compliance specialists appears to almost triple for firms between 50 and 5,000 employees. We also verify that the evidence of increasing specialization in regulatory compliance is present across all different industries and it is not driven solely by certain sectors, such as finance or utilities.

4.2 Size-dependent regulatory requirements and enforcement

As discussed in [Section 2](#), the share of regulatory compliance costs RegIndex_{it} may be driven by both the extent of regulatory agencies' regulatory requirements, $\tilde{R}_{it}$, and endogenous compliance due to enforcement effort, E_{it} (which affects the likelihood of inspection p_{it}). In particular,

from Equations (1) and (2), we know $\text{RegIndex}_{it} = \frac{(w_{it}^* p_{it} \tilde{R}_{it})^{0.5}}{w_{it}}$, where $\tilde{R}_{it}$ captures the intensity of regulation requirements and p_{it} captures the enforcement stringency of agencies. These competing drivers, which may be size-dependent, introduce a challenge in interpreting the findings in [Section 3](#). For example, a small firm may exhibit low regulatory compliance costs because the agency rules are designed to be lighter on small firms

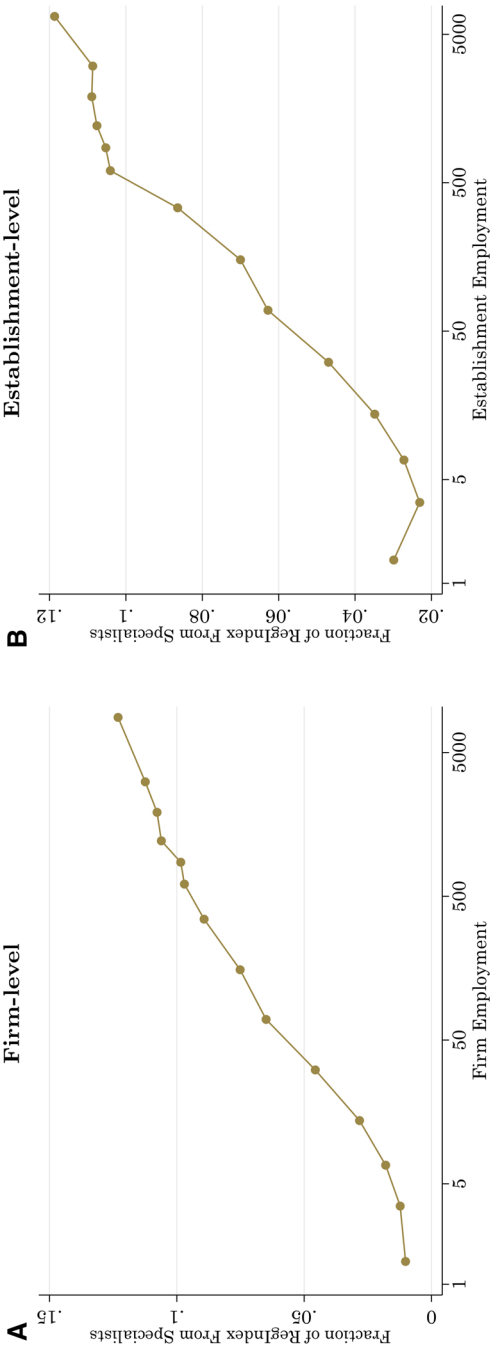


Figure 11
Regulatory compliance specialization and size

This figure plots the fraction of RegIndex due to compliance specialists by employment bins for firms in panel A and establishments in panel B. Compliance specialists are defined as the top 25 occupations with the highest regulation intensity in Table IA.2. We decompose a business's compliance costs into those paid to specialists and those paid to non-specialists. We plot the average fraction from specialists in businesses from each employment bin, where the bins are [1, 2], [3, 4], [5, 9], [10, 19], [20, 49], [50, 99], [100, 249], [250, 499], [500, 749], [750, 999], [1000, 1499], [1500, 2499], [2500, 3999], and above 4000. The x-axis is on a log scale.

(proxied by lower $\bar{R}_{it}$) or because small firms comply less, as they know they are more likely to fly under the radar of inspectors (proxied by lower p_{it} for smaller firms). These considerations are relevant to decomposing the non-monotonic shape of the relationship between compliance costs and firm size in [Section 3](#).

We present a methodology to analyze the effect of regulation requirements and enforcement on the inverted-U relation between RegIndex and firm size. Our approach takes two steps. First, we use changes in regulation requirements and enforcement to estimate each regulation-related task's sensitivity to requirements and to enforcement.⁵³ Second, we construct two sub-components of RegIndex: the share of a firm's labor spending on requirement-sensitive tasks, $\text{RegIndex}_{i,t}^{req}$, and the share of a firm's labor spending on enforcement-sensitive tasks, $\text{RegIndex}_{i,t}^{enf}$. This decomposition is constructed so that $\text{RegIndex}_{i,t} = \text{RegIndex}_{i,t}^{req} + \text{RegIndex}_{i,t}^{enf}$. We then analyze the role of each subcomponent in driving our main results.

4.2.1 Methodology for estimating task sensitivity to requirements and enforcement.

Assume that certain regulation-related tasks may be more relevant for complying with regulatory requirements, while other regulation-related tasks may be more relevant for dealing with enforcement. Firms respond to changes in regulatory requirements ($\bar{R}_{it}$) by adjusting spending on requirement-sensitive tasks and respond to changes in enforcement (p_{it}) by adjusting spending on enforcement-sensitive tasks.⁵⁴ To estimate each regulation-related task's sensitivity to regulation requirements and enforcement, we develop a methodology based on a shift-share approach.

For each regulatory agency k at time t , suppose being able to measure both its regulatory requirements, reg_{kt} , and the extent of the enforcement and supervision effort, enf_{kt} . Further, suppose we are also able to measure industry j 's regulation originating from agency k at time t , r_{jkt} , that is, the industry-specific exposure to each regulatory agency.⁵⁵ How to perform these operations is discussed below in [Section 4.2.2](#).

⁵³ For instance, one would expect the task to "... learn about and analyze the potential impacts of proposed environmental policy regulations" (O*Net task ID 19529) would be more sensitive to new regulatory requirement, while the task to "... coordinate investigative efforts with law enforcement officers and attorneys" (O*Net task ID 16055) would be more sensitive to regulatory enforcement.

⁵⁴ Some regulation-related tasks will likely be sensitive to both requirements and enforcement. Our estimation allows for such "mixed" tasks.

⁵⁵ We define the agency exposure at the industry level rather than firm level due to an empirical limitation. The only information we possess that can inform firms' exposure to regulatory agencies is the regulation-related tasks they perform. A nontrivial amount of small firms in our data do not perform regulation-related tasks (consistent with our inverted-U finding). However, this does not mean that the small firms are not exposed to regulatory agencies' future changes in regulation requirements and enforcement. Defining the agency

It is possible to create two shift-share variables tracing changes in regulatory requirements and in enforcement pertinent to firm i in industry j as the interaction of shifts and shares:

$$iv(\Delta \log(\tilde{R}_{it})) = \sum_k \Delta \log reg_{kt} \times r_{j(i)kt}$$

$$iv(\Delta \log(p_{it})) = \sum_k \Delta \log enf_{kt} \times r_{j(i)kt}, \quad (5)$$

where $\Delta \log reg_{kt}$ and $\Delta \log enf_{kt}$ are the agency-specific requirement (log) changes and enforcement changes from year t to $t+3$, and $r_{j(i)kt}$ is industry j 's exposure to agency k 's regulations in year t .

One can now categorize each regulation-related task o as *requirement-sensitive* or *enforcement-sensitive* by estimating the following regression for each task o :

$$\Delta \log(\text{RegIndex}_{i,t}) = \alpha_o + \beta_o iv(\Delta \log(\tilde{R}_{it})) + \gamma_o iv(\Delta \log(p_{it})) + \delta_o X_{it} + \xi_{it}, \quad (6)$$

where $\Delta \log(\text{RegIndex}_{i,t})$ is firm i 's change in spending on task o from year t to $t+3$.⁵⁶ In Equation (6) β_o and γ_o , respectively, measure task o 's sensitivity to regulatory requirement changes and to enforcement changes.⁵⁷

In practice, we run the regression (6) for each regulation-related task o on standardized shift-shares and categorize a task as follows: If $\beta_o > \gamma_o$ and is statistically significant at the 5% level, we regard task o as requirement-sensitive. If $\beta_o < \gamma_o$ and is significant at the 5% level, we regard task o as enforcement-sensitive. Finally, if the two coefficients are not statistically different at five percent significance level, we regard task o as to be mixed (half requirement-sensitive and half enforcement-sensitive).⁵⁸

exposure at the industry level ensures that we capture all firms' exposure to future regulatory changes, despite the fact that they can have different responses to the future changes, as we show in Section 4.2.3.

⁵⁶ Our empirical implementation adopts the Haltiwanger percentage growth rate to proxy for the log change, $\Delta \log(\text{RegIndex}_{i,t}) \cong \frac{\text{RegIndex}_{i,t+3} - \text{RegIndex}_{i,t}}{(\text{RegIndex}_{i,t+3} + \text{RegIndex}_{i,t})/2}$, to avoid taking logarithms of 0 (consider, e.g., a firm that changed from zero spending on task o to a positive spending on task o in response to regulatory changes). The Haltiwanger growth rate is 98% correlated with log changes in RegIndex in observations where both measures are available. We choose the 3-year horizon for all log changes in Equation (6), as well as the two IVs, because establishments are covered by the OEWS sample every 3 years.

⁵⁷ We also include firm-level controls X_{it} , firms' initial RegIndex level to account firm heterogeneity; year fixed effects to account for macro conditions; and changes in firms' total labor costs and in the average wage rate of firms' regulation-related tasks, which are motivated by the framework in Section 2.2. In particular, recall that plugging Equation (1) into Equation (2) and taking log changes, follows $\Delta \log(\text{RegIndex}_{i,t}) = \frac{1}{2} \Delta \log(\tilde{R}_{it}) + \frac{1}{2} \Delta \log(p_{it}) + \frac{1}{2} \Delta \log(w'_{it}) - \Delta \log W_{it}$.

⁵⁸ We also conduct robustness analyses using 10% or 1% significance level to label tasks. The results are qualitatively similar and available from the authors upon request.

Following this procedure, our estimation below shows that 49% of regulation-related tasks are labeled as requirement-sensitive, 16% as enforcement-sensitive, and 35% as mixed. See Table IA.8 for examples. Once each task is categorized, we next adopt the exact definition in Equation (2) to create a requirement-specific $\text{RegIndex}_{i,t}^{req}$ by aggregating only the percentage costs for performing the requirement-sensitive tasks with the firm (establishment), and an enforcement-specific $\text{RegIndex}_{i,t}^{enf}$ by aggregating only the percentage costs for performing the enforcement-sensitive tasks.

4.2.2 Empirical implementation. Let us now come back to the empirical components of the two variables in Equation (5): each regulatory agency's regulatory requirement changes, $\Delta \log \text{reg}_{kt}$; enforcement changes, $\Delta \log \text{enf}_{kt}$; and each NAICS 6-digit industry's exposure to each regulatory agency, r_{jkt} . We select 12 major regulatory agencies for which we can measure changes in regulatory requirements and enforcement. These agencies together account for 82% of the non-Treasury regulatory compliance burden (by hours) in the United States during 2002–2014. See all details for the construction in the Internet Appendix B.

We measure changes in each major agency k 's regulatory requirements, $\Delta \log \text{reg}_{kt}$, based on changes in regulations from agency k from t to $t+3$. For each fiscal year, major agencies need their estimates of the changes in regulatory compliance hours for regulations under their oversight with the OIRA (Kalmenovitz 2023). We use 3-year log differences in reported compliance hours due to changes in enactment and retirement of regulations to measure regulation-requirement changes of the agency.⁵⁹

We measure changes in each major agency k 's enforcement, $\Delta \log \text{enf}_{kt}$, using panel data of U.S. federal government employees from 2002 to 2014.⁶⁰ This individual-level database is compiled by the U.S. Office of Personnel Management's (OPM) Enterprise Human Resources Integration System. The data are made available by BuzzFeed News through a Freedom of Information Act request. The data cover detailed information of all federal employees, except the Department of Defense, at a quarterly frequency.⁶¹ Variables crucial for our study include the employee's agency, occupation,

⁵⁹ To account for regulatory changes only due to enactment and retirement of regulations, we exclude changes in regulatory compliance hours due to agencies' re-estimation of the compliance hours for the existing regulations in our calculation. Data can be downloaded from the "Information Collection Budget of the United States Government" reports at <https://www.whitehouse.gov/omb/information-regulatory-affairs/reports/#ICB>.

⁶⁰ See Levinson (1996); Jackson and Roe (2009); Jackson (2007); Muehlenbachs, Staubli, and Cohen (2016) for examples of prior studies using regulatory agencies' employment to measure their enforcement efforts.

⁶¹ The data includes 206 million observations and can be downloaded at <https://archive.org/download/opm-federal-employment-data>. We thank Joe Raffiee for introducing this data to us.

and full-time/part-time employment status.⁶² We identify each occupation as “regulation-related” using the same list of keywords and procedure in Section 2. We use 3-year log differences in regulation-related full-time employment in each agency to measure enforcement changes of the agency.

Figure IA.13 traces the time-series of the two changes ($\Delta \log reg_{kt}$ and $\Delta \log enf_{kt}$) for each of the 12 regulatory agencies. We observe that there is substantial independent variation in the two change series across all regulators, although for some agencies the separation is starker. Our estimation procedure in Equation (6) exploits the independent variations of the two changes to assign whether a task is requirement-sensitive or enforcement-sensitive by running regressions that include both variables.

The shift-share variables in Equation (5) also require measuring each industry’s exposure to regulatory agencies’ changes, r_{jkt} . We compute this exposure based on the textual similarity between the regulation-related tasks that the industry performs and the rules that agency’s regulation specifies. This approach is similar to Al-Ubaydli and McLaughlin (2017), who measure the probability that a given regulatory restriction targets a specific industry based on industry descriptions and Armstrong, Glaeser, and Hoopes (2025), who measure publicly traded firms’ exposure to various government agencies based on their 10-K filings. Specifically, the approach has two steps. First, we extract the top 50 identifying keywords for each regulatory agency using natural language processing of the Code of Federal Regulation text,⁶³ and we compute a Google BERT similarity between each regulation-related task o and an agency k ’s keywords, $\phi_{k,o}$.⁶⁴ We standardize the 12 similarities for each task to sum up to one, that is, $\sum_k \phi_{k,o} = 1$. These standardized similarities capture a regulation-related

⁶² Importantly, while the data adopt a different occupation classification system from SOC, we are able to obtain each federal employee occupation’s task description from the “Handbook of Occupational Group and Families” on the OPM website. The OPM has its own definitions for government occupations that are different from the SOC system. The handbook for OPM occupation description can be downloaded at <https://www.opm.gov/policy-data-oversight/classification-qualifications/classifying-general-schedule-positions/occupationalhandbook.pdf>.

⁶³ We obtain the keywords in three steps: First, we manually select all the CFR chapters that mentions the names of the 12 agencies and their subagencies, where the CFR texts are available at <https://www.govinfo.gov/app/collection/cfr/2021/>. For instance, the CFR chapters that mention Environmental Protection Agency are Title 40 Chapter I (Part 1–1099) and Chapter IV (Part 1400–1499). Second, we count the term frequency of each word in the overall selected CFR text ($count_{all}$) and also in the texts about each agency k ($count_k$). We identify keywords as specific to agency k if the keywords appear over 50% of the time in agency k ’s texts alone, that is, $count_k / count_{all} \geq 0.5$. Third, we remove uninformative words from the list, such as abbreviations by computing the similarity of the keywords to its agency’s name using Google Bidirectional Encoder Representations from Transformers (BERT). Our final keywords are the top 50 keywords that have the highest BERT similarity score. Table IA.9 lists the 50 keywords for each of the 12 agencies.

⁶⁴ Google BERT similarity captures contextual and semantic meaning in text by leveraging deep language representations, making it more accurate for understanding nuanced or ambiguous language compared to standard cosine similarity. In contrast, standard cosine similarity relies on static vector representations and often fails to account for word order or context, limiting its effectiveness in complex linguistic tasks.

task's exposure to the 12 agencies.⁶⁵ Second, we have each industry j 's percentage labor spending on each regulation-related task o , $s_{j,o,t}$. Hence, we compute industry j 's beginning-of-period exposure to agency k 's regulations at t (r_{jkt} in Equation (5)) as the weighted average of its regulation-related task's exposure to agency k , that is, $r_{jkt} = \sum_o s_{j,o,t} \times \phi_{k,o}$. With all elements in Equation (5) now available, we can construct $iv(\Delta \log(p_{it}))$ and $iv(\Delta \log(\tilde{R}_{it}))$. Applying the two shift-share variables to Equation (6), we obtain each regulation-related task's sensitivities to regulatory requirement changes and enforcement changes, which ultimately helps us label each task as requirement-sensitive, enforcement-sensitive, or both following the methodology in Section 4.2.1. Once each regulation-related task is labeled, we decompose a firm's (establishment's) RegIndex into $\text{RegIndex}_{i,t}^{req}$ and $\text{RegIndex}_{i,t}^{enf}$.

4.2.3 Results of decomposing RegIndex.

Decomposing RegIndex in Levels. Figure 12 plots the two components of RegIndex, RegIndex^{req} and RegIndex^{enf} for different size bins. Panel A plots the results for firms, and panel B for establishments. For each employment bin, we observe the average share of the total wage bill of a firm or of an establishment associated with performing compliance tasks related to requirements and of those related to enforcement.

The figure demonstrates how, for both firms and establishments, the inverted-U relationship between RegIndex and size originally observed in Figure 5 can be primarily traced to regulatory requirements, that is, RegIndex^{req} . Firms in middle-sized bins around 500 employees have 0.57% higher RegIndex than those in the smallest bin (1.67%–1.1%). The differences in RegIndex^{req} between the two groups of firms is 0.55%, accounting for 97% ($=0.55/0.57$) of the difference in RegIndex. In contrast, the relation between RegIndex^{enf} and firm size appears flat. Indeed, regulatory requirements, RegIndex^{req} , account for almost the entire difference in RegIndex between mid-sized firms and the largest size group.

Overall, Figure 12 does not support that differential enforcement for large and small firms as a quantitatively important mechanism behind the inverted-U shape relationship between RegIndex and firm size. Heterogeneous regulatory requirements on small and large firms appear to drive essentially the entirety of the size-dependence of regulatory compliance costs incidence. This is consistent with the pervasive implementation of threshold-based regulations in practice, commonly known as regulatory

⁶⁵ See Chen and Kalmenovitz (2024) for an application of a supervised machine-learning algorithm to compute the regulatory similarity of each pair of S&P1500 firms.

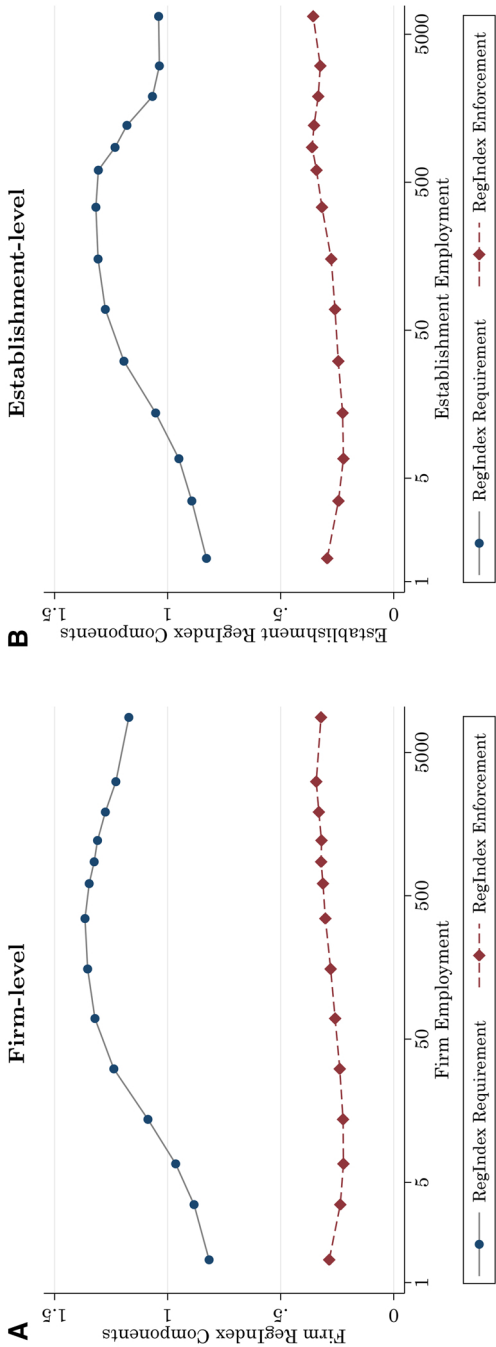


Figure 12
Decomposing RegIndex: Regulatory requirements versus enforcement

This figure plots the two components of RegIndex as firms' regulatory compliance due to regulatory requirements and due to regulatory enforcement. See [Section 4](#) for details of the decomposition procedure. We plot the average RegIndex for regulatory requirements and enforcement, respectively, from each employment bin, where the bins are [1, 2], [3, 4], [5, 9], [10, 19], [20, 49], [50, 99], [100, 249], [250, 499], [500, 749], [750, 999], [1000, 1499], [1500, 2499], [2500, 3999], and above 4000. The x-axis is on a log scale.

tiering. In Table IA.18, we present a varied sample of regulatory thresholds in firm size. As can be seen, size thresholds in employment, revenues or assets are common in regulations covering a broad spectrum of industries (e.g., Agriculture, Finance and Insurance, Manufacturing, Retail, Mining, and Utilities) both on a federal and state level.

Sensitivity to Requirements and Enforcement Changes. We can further corroborate the above decomposition of RegIndex in levels with an analysis of RegIndex's changes. Consider the regression of (log) changes in RegIndex on the regulatory requirements and on the enforcement shift-shares:

$$\Delta \log(\text{RegIndex}_{it}) = \delta_0 + \beta_1 iv(\Delta \log(\tilde{R}_{it})) + \gamma_1 iv(\Delta \log(p_{it})) + \delta_1 X_{it} + \xi_{it}, \quad (7)$$

where all variables are as in Equation (6).

Panel A of Table 6 reports our estimation of Equation (7) for the full sample of firms and establishments. For comparability, coefficients are reported for standardized variables.

We begin by investigating the roles of $iv(\Delta \log(p_{it}))$ and $iv(\Delta \log(\tilde{R}_{it}))$ separately. Columns (1) and (4) examine the response of firms' and establishments' share of regulatory costs ($\Delta \log(\text{RegIndex}_{it})$) to enforcement changes, $iv(\Delta \log(p_{it}))$. In both columns, we observe positive and statistically significant coefficients, confirming the intuitive conditional correlation between increases in agency regulatory hires translating into more enforcement and, consequently, higher regulatory compliance expenditure. The estimated coefficients further indicate a quantitatively meaningful relationship, as a one standard deviation increase in $iv(\Delta \log(p_{it}))$ produces a 0.21 of a standard deviation increase in the change in regulatory compliance costs index for a firm (0.23 for an establishment). Columns (2) and (5) examine the response of firms' and establishments' regulatory compliance costs to regulation requirements changes, $iv(\Delta \log(\tilde{R}_{it}))$. The coefficient for $iv(\Delta \log(\tilde{R}_{it}))$ is larger than for the case of $iv(\Delta \log(p_{it}))$. A one standard deviation increase in $iv(\Delta \log(\tilde{R}_{it}))$ produces a 0.26 of a standard deviation increase in the change in regulatory compliance costs index for a firm (0.27 for an establishment). These results suggest that our shift-shares meaningfully correspond to ex post changes in firms' regulatory compliance costs.

Columns (3) and (6) include both requirements and enforcement variables in the estimation of Equation (7), allowing us to compare the marginal effect of changes in each component. Confirming our results above, this horse-race specification reveals a stronger role for regulatory requirements than enforcement in driving changes in compliance. The regulatory requirements coefficient is estimated at 0.23 and statistically

Table 6
Response of RegIndex to regulatory requirement and enforcement changes
A: Full sample

	Firm level			Establishment level		
	(1)	(2)	(3)	(4)	(5)	(6)
$iv(\Delta \log(p_{it}))$	0.209** (0.077)		0.112 (0.075)	0.234*** (0.063)		0.128* (0.065)
$iv(\Delta \log(\bar{R}_{it}))$		0.257*** (0.038)	0.228*** (0.054)		0.269*** (0.036)	0.231*** (0.052)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	608,500	608,500	608,500	628,733	628,733	628,733
Adjusted R^2	0.322	0.340	0.344	0.323	0.340	0.345

B: Subsamples by size

	Firm level				Establishment level			
	1–19 (1)	20–399 (2)	400–749 (3)	≥ 750 (4)	1–19 (5)	20–399 (6)	400–749 (7)	≥ 750 (8)
$iv(\Delta \log(p_{it}))$	0.102 (0.075)	0.120 (0.078)	0.099 (0.076)	0.083 (0.064)	0.115* (0.062)	0.140* (0.066)	0.168** (0.074)	0.077 (0.066)
$iv(\Delta \log(\bar{R}_{it}))$	0.169** (0.056)	0.258*** (0.054)	0.289*** (0.054)	0.309*** (0.046)	0.170** (0.053)	0.271*** (0.050)	0.313*** (0.053)	0.367*** (0.055)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	189,404	352,779	29,482	36,835	220,464	375,485	19,622	13,162
Adjusted R^2	0.400	0.324	0.243	0.214	0.397	0.321	0.265	0.233

This table reports the results of regressing 3-year changes in a firm's log RegIndex on the shift-share variables. Panel A shows the results using the full sample, and panel B shows the results in subsamples divided by firms or establishments' employment size. Equation (7) provides the regression specification. $iv(\Delta \log(p_{it}))$ and $iv(\Delta \log(\bar{R}_{it}))$ are the shift-share variables for the NAICS 6-digit industry's enforcement changes and regulatory-requirement changes. Section 4 provides details on the construction of the shift-share variables. All regressions include controls of 3-year changes in the log total wage costs of the firm or establishment, 3-year changes in the log wage rate of the regulation-related tasks, and beginning of period log RegIndex to account for mean-reversion of firms' or establishments' RegIndex, and year fixed effects. All variables are standardized to have mean 0 and variance of 1 for ease of interpretation. Standard errors are double-clustered by industry and year and presented in parentheses. The sample period is from 2002 to 2011. * $p < .1$; ** $p < .05$; *** $p < .01$.

significant, while the enforcement effort slope is 0.11 and statistically insignificant for firms (similarly for establishments).⁶⁶

To see how the shift-share slopes may be informative about the mechanisms behind the relation between the level of RegIndex and business size, divide firms in four employment bins: between 1 and 19, 20 and 399, 400 and 749, and above 750 employees. Panel B of Table 6 confirms that the effects of changes in regulatory requirements, $iv(\Delta \log(\bar{R}_{it}))$, are substantially higher for medium and large firms relative to small ones, particularly for those entities employing ≤ 10 workers. The null hypotheses of equality of the coefficients on the variable $iv(\Delta \log(\bar{R}_{it}))$ for firms below

⁶⁶ Table 1A.12 displays variations of the above results across broad sectors of firms and establishments.

10 employees and for those between 400–749 or those above 749 employees are rejected at standard levels of statistical significance. This is consistent with an intuitive mechanism. Under regulatory tiering, which implies less stringent regulatory requirements for small business, such as in the cases of local banks or labor laws applying to firms below certain employment thresholds, changes in requirements affecting medium and large firms have only indirect or general equilibrium effects, as small businesses are exempt. This in turn explains the smaller coefficient on $iv(\Delta \log(\tilde{R}_{it}))$. Notice that, as regulatory tiering is a feature of the design of regulatory requirements, it is important to hold enforcement of regulation constant, to cleanly isolate this mechanism in the tests of Table 6.⁶⁷

In summary, the evidence from decomposing the level of firm RegIndex and inspecting the sensitivity of RegIndex changes to requirement and enforcement changes suggests that regulatory requirements and regulatory tiering seem to be important drivers of the inverted-U relationship between firms' RegIndex and their employment size.

4.3 Potential implications for firm size distribution

Potential distortions to the establishment and firm size distributions ensue from the presence of the size-dependent regulation that we just documented.⁶⁸ To precisely quantify aggregate macroeconomic implications, however, one would need to decouple the role of regulatory requirements and endogenous enforcement from concomitant market forces (e.g., technology or international trade) and unobserved sector-specific shocks. In this study, we do not possess an appropriate identification strategy to pursue this rigorously and any evidence in terms of macroeconomic implications should be taken as exploratory, rather than conclusive. Nonetheless, one can outline some implications useful to motivate future research.

Our simple theoretical framework in Section 2.2 suggests that firms choose their optimal size in an environment where regulatory compliance

costs are size-dependent, that is, $L_{it}^* = \left(\frac{\phi_l - w_{it}}{(\alpha + \beta)(w_{it}^r \pi \rho)^{\frac{1}{2}}} \right)^{\frac{2}{\alpha + \beta - 2}}$ in Equation

(1). This, in turn, implies that policy parameters $(\alpha, \beta, \rho, \pi)$ directly affect the optimal firm size distribution. We thus can offer a discussion of the association between the size-dependent regulatory compliance costs and

⁶⁷ This is why controlling for $iv(\Delta \log(p_{it}))$, which may also be size-dependent in principle, is a crucial step in isolating the role of regulatory tiering. For recent research using regulatory tiering for statistical identification of regulatory distortions, see Garicano, Lelarge, and Van Reenen (2016); Aghion, Bergeaud, and Van Reenen (2023); Alvero, Ando, and Xiao (2023); Ewens, Xiao, and Xu (2024).

⁶⁸ For the case of French labor regulation, see Aghion, Bergeaud, and Van Reenen (2023).

the dynamics of firm/establishment size distribution in the United States from 2002 to 2014.

We begin by performing the simple exercise of examining the conditional correlation between changes in firm size distribution and *RegIndex* within the sector. To account for changes in sectoral composition in the economy, we estimate the following regression:

$$\Delta FirmShare_{l,j} = \varphi RegIndex_{l,j} + \delta_j + \epsilon_{l,j},$$

where $\Delta FirmShare_{l,j}$ measures the change in the mass of firms from each of 14 different employment bins (l) within each of eight different sectors (j) between the years 2002 and 2014, $RegIndex_{l,j}$ is the average *RegIndex* for the employment bin within the sector (see Figure 6), and δ_j represents sector fixed effects. Intuitively, in this regression a negative φ estimate indicates a “hollowing out” of an employment bin within the sector whenever the level of *RegIndex* in that employment-sector bin is higher.⁶⁹

Indeed, panel A of Table IA.19 reports a negative and statistically significant estimate of $\varphi = -0.017$ ($t = -3.16$) for changes in firm size distribution (see column (1)). The estimates are similar for changes in establishments’ size distribution $\varphi = -0.016$ ($t = -3.67$), as shown in panel B. The economic magnitude is sizable, though we emphasize that these results should not be interpreted causally: size bins with a 1-standard-deviation higher *RegIndex* within a sector are associated with a 0.7-standard-deviation larger loss in firm or establishment share. Columns (2) and (3) further split the sample into the smaller and larger halves of the firm size distribution, revealing that the effects are concentrated among small and mid-sized firms rather than large firms. This finding aligns with the view that threshold-based regulations tend to benefit small firms while hollowing out mid-sized firms, since most U.S. regulatory thresholds fall within this range (see examples in Table IA.18); correspondingly, *RegIndex* is consistently lower for small firms than for mid-sized firms across nearly all sectors (Figure 6).

To further confirm the negative association between regulatory compliance burden and firm size distribution, in columns (4)–(6) of Table

⁶⁹ To measure $\Delta FirmShare_{l,j}$, we first compute the share of establishments in each of the 14 size bins (Section 3) within each NAICS 1-digit sector in each year. To ensure the shares are representative of the size distribution in the economy, we weigh each establishment by its sampling weights assigned by the BLS—BLS assigns sampling weights to establishments by size, location, industry, and ownership so as to produce the aggregate statistics at the industry, geographic, and national level for public use at <https://www.bls.gov/oes/tables.htm>. Next, we compute the changes in the share of establishments in each employment-sector bin from 2002 to 2014. We also compute the changes in firms’ size distribution using the same methodology. For single-establishment firms, their sampling weights are the same as their establishments’. Assigning sampling weights to multi-establishment firms is challenging. We use the lowest sampling weights among establishments within the firm, so as to mitigate the impact of large sampling weights due to low coverage of businesses in that establishment’s sampling bucket. Nevertheless, the results are very similar if we use the average weight of establishments within the firm as the firm’s weight.

IA.19, we run another regression specification based on changes in the employment-sector bin's RegIndex from 2002 to 2014:

$$\Delta FirmShare_{l,j} = \varphi \Delta RegIndex_{l,j} + \delta_j + \epsilon_{l,j}.$$

In this regression, φ estimates the conditional correlation between growth in the mass of firms in an employment bin and the increase in compliance costs in that bin. Consistent with our finding above, we observe a significantly negative estimate of φ in column (4). The economic magnitude, measured in one-standard-deviation terms, is less than half that of the RegIndex level in column (1). In columns (5) and (6), we further separate the sample into the smaller and larger halves of the firm size distribution. Again, we observe that the results are driven by changes in regulation affecting small and mid-sized firms. In other words, this evidence suggests that small firms' growth is also more sensitive to changes in regulatory costs—another relevant dimension when assessing the implications of rising compliance costs for firm dynamism. Our RegIndex, which accounts for all firms—including the very smallest—is thus particularly valuable for future research on this topic.

There is a direct implication of these correlation estimates when combined with the lower regulatory burden falling on the very small firms and the higher burden on mid-sized firms at the aggregate level shown in Figure 6: a loss of mass in the middle of the firm (and establishment) size distribution over time. We observe some suggestive evidence of this phenomenon in Figure IA.15. The figure plots the changes in firms' and establishments' size distribution from 2002 to 2014 in the U.S. economy from our microdata. We observe that the fraction of small and mid-sized businesses with employees between 10 and 250 declines relative to the very small or very large firms, consistent with the intuition that mid-sized firms face higher regulatory compliance costs. The biggest expansion over 2002–2014 occurs for the very small firms, where RegIndex is the lowest. While this pattern should not be uniquely ascribed to regulatory requirements, as many omitted confounding factors may affect this negative correlation, it provides suggestive evidence of the potentially broader implications of our work.

5. Conclusion

This paper presents a new approach to estimating business compliance costs of regulation in the United States. Based on micro establishment-occupation-level data, we quantify the total labor costs paid by businesses to employees engaging in safety, compliance, monitoring, and other regulation-related tasks in order to meet federal, state, and local regulatory requirements. The average U.S. firm spends between 1.3% and 3.3% of its total wage bill on regulatory compliance. This wage bill grew at an

annual rate of about 1% from 2002 to 2014, roughly half of the average annual GDP growth rate over the period. The total wage bill devoted to regulatory compliance workers in 2014 was between \$79 billion and \$239 billion, depending on the stringency of the regulatory compliance measure employed, and up to \$289 billion when capital is also added, comparable to U.S. gross business income taxes.

We show that the percentage of firms' labor costs paid for regulatory compliance first increases with firm size, measured by total employment, and then decreases, exhibiting an inverted-U shape. This inverted-U relationship is a robust pattern, as we show across multiple specifications—when accounting for within-occupation variation in regulatory tasks, incorporating capital expenditures, adjusting for outsourcing of compliance functions, and controlling for multi-state regulatory exposure. These consistent findings strengthen our confidence in the reliability of the inverted-U pattern as a structural feature of regulatory burden distribution.

This inverted-U shaped relationship between RegIndex and firm size suggests that, for small businesses, regulation is tiered and tends to be lighter, while red tape increases as employment reaches 500 workers. Beyond this threshold, regulatory costs tend to decrease, with evidence of economies of scale kicking in for regulatory compliance. Identifying the presence of increasing returns to scale in regulatory compliance for medium-large firms is an important step in the direction of assessing equilibrium firm size distortions due to the design of the U.S. administrative system. We further design and implement a shift-share method to identify firm responses to regulation requirements versus enforcement. Using this design, we argue that changes in regulatory requirements appear to contribute significantly to the enhanced inverted-U relationship between firms' regulatory compliance cost and size in our sample period, while the contribution from changes in enforcement is limited.

Future research should extend the use of our methodology to other high-income countries, where similar microdata is available, for a comparative perspective on the costs of regulation and assessing external validity. Quantifying the productivity losses to U.S. firms due to regulation also appears to be an important direction of future inquiry.

Code Availability: The replication code is available in the Harvard Dataverse at <https://doi.org/10.7910/DVN/ER5UBH>.

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